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Ohashi et al.

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(54) **SEMICONDUCTOR DEVICE, INVERTER
CIRCUIT, DRIVING DEVICE, VEHICLE,
AND ELEVATOR**

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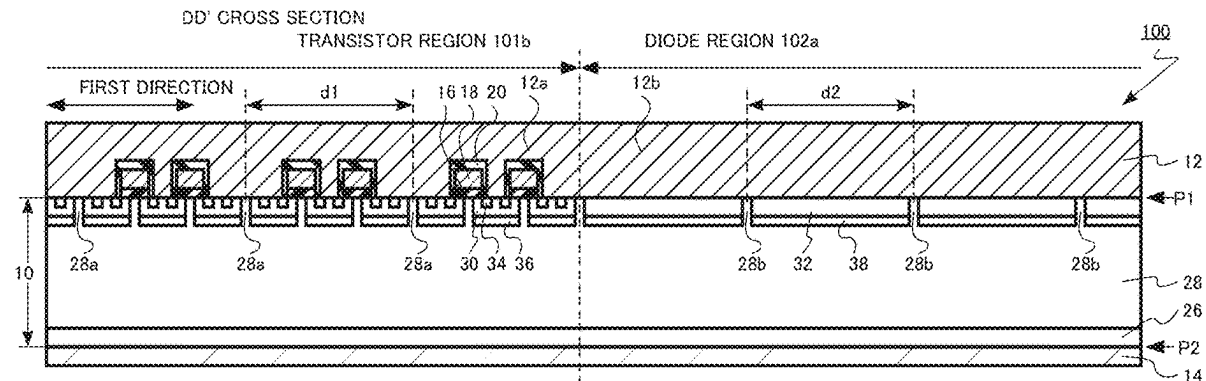
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(57) **ABSTRACT**

A semiconductor device according to an embodiment includes a semiconductor chip having a transistor region and a diode region, and a conductor. The semiconductor chip includes a first electrode, a second electrode, a silicon carbide layer between the first electrode and the second electrode, and a gate electrode. The first electrode includes a first region in the transistor region and a second region in the diode region. A first contact area between the conductor and the first region is larger than a second contact area between the conductor and the second region.

13 Claims, 24 Drawing Sheets



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B60R 1/02 (2006.01)
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B66B 11/04 (2006.01)
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- (52) **U.S. Cl.**
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(2013.01); *H02P 27/06* (2013.01)
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FIG.1A

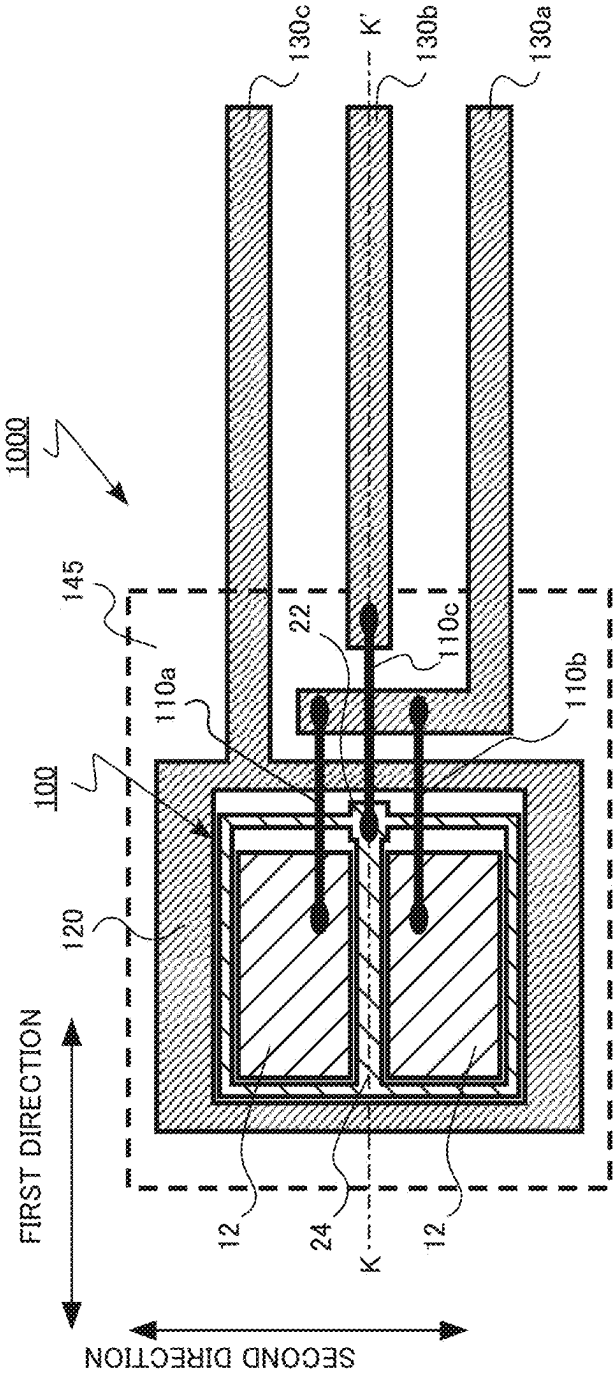


FIG.1B

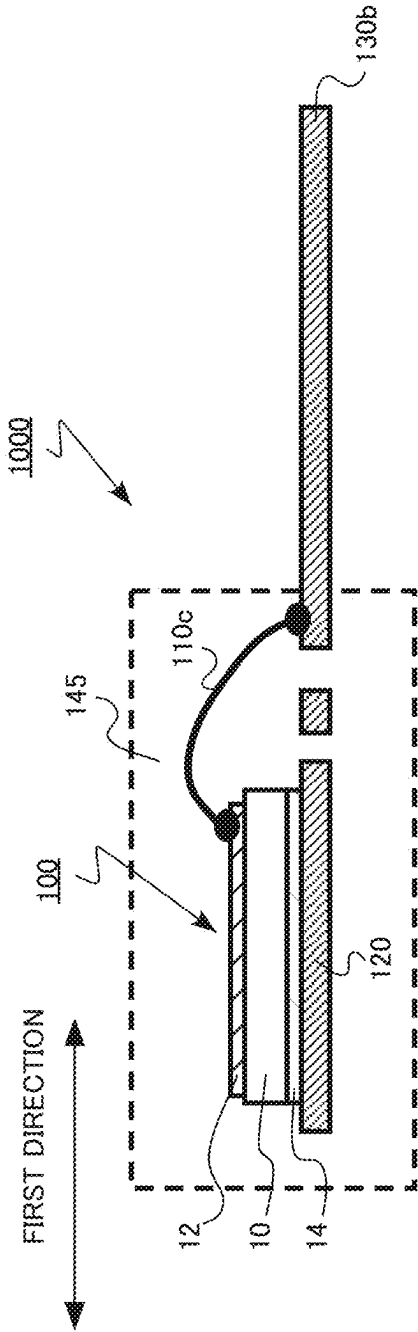


FIG.2A

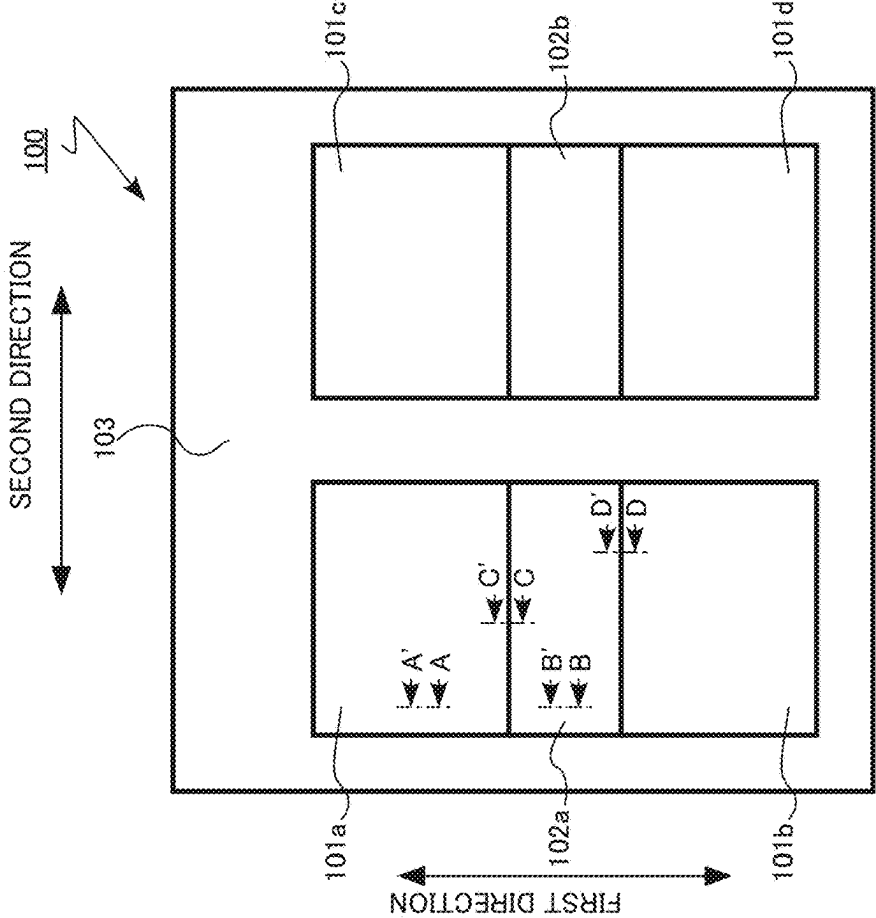
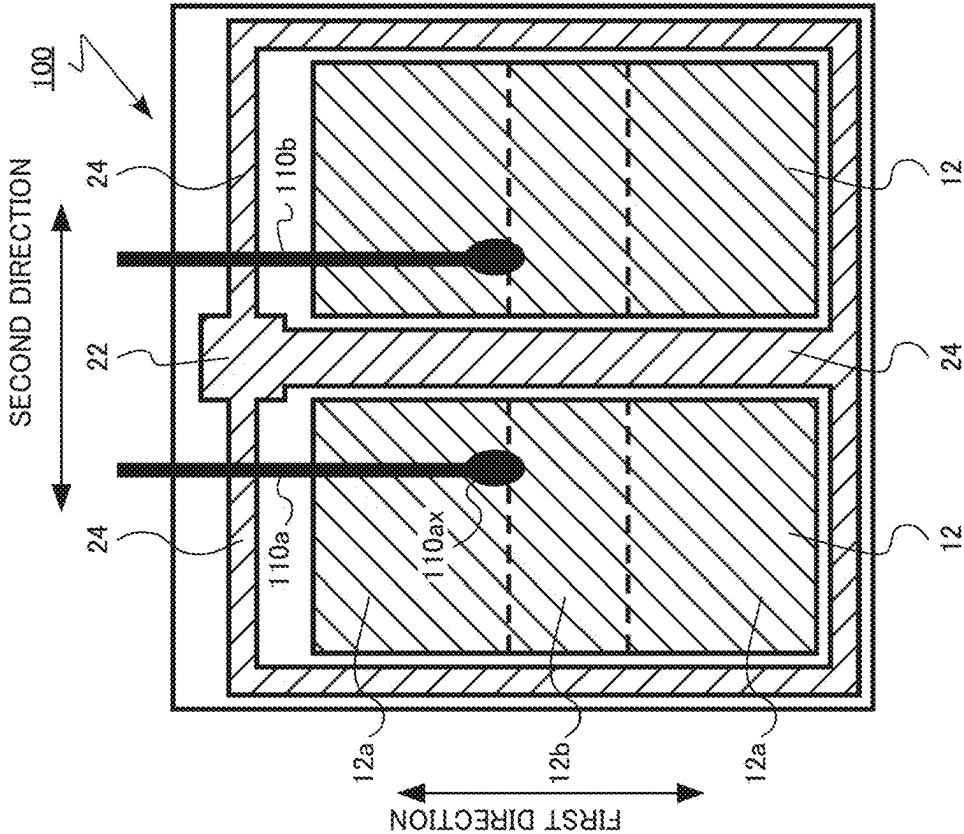


FIG.2B



364

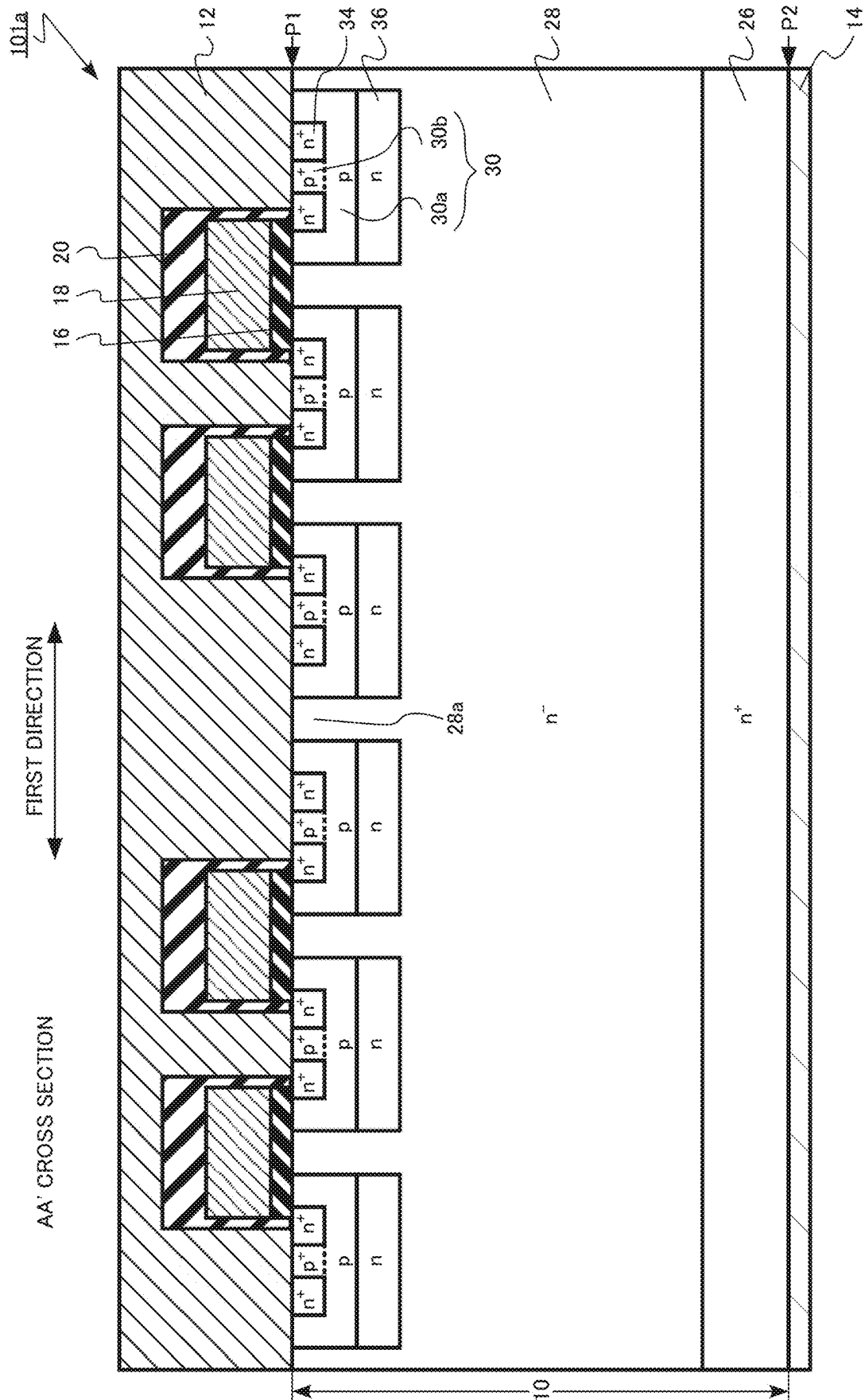
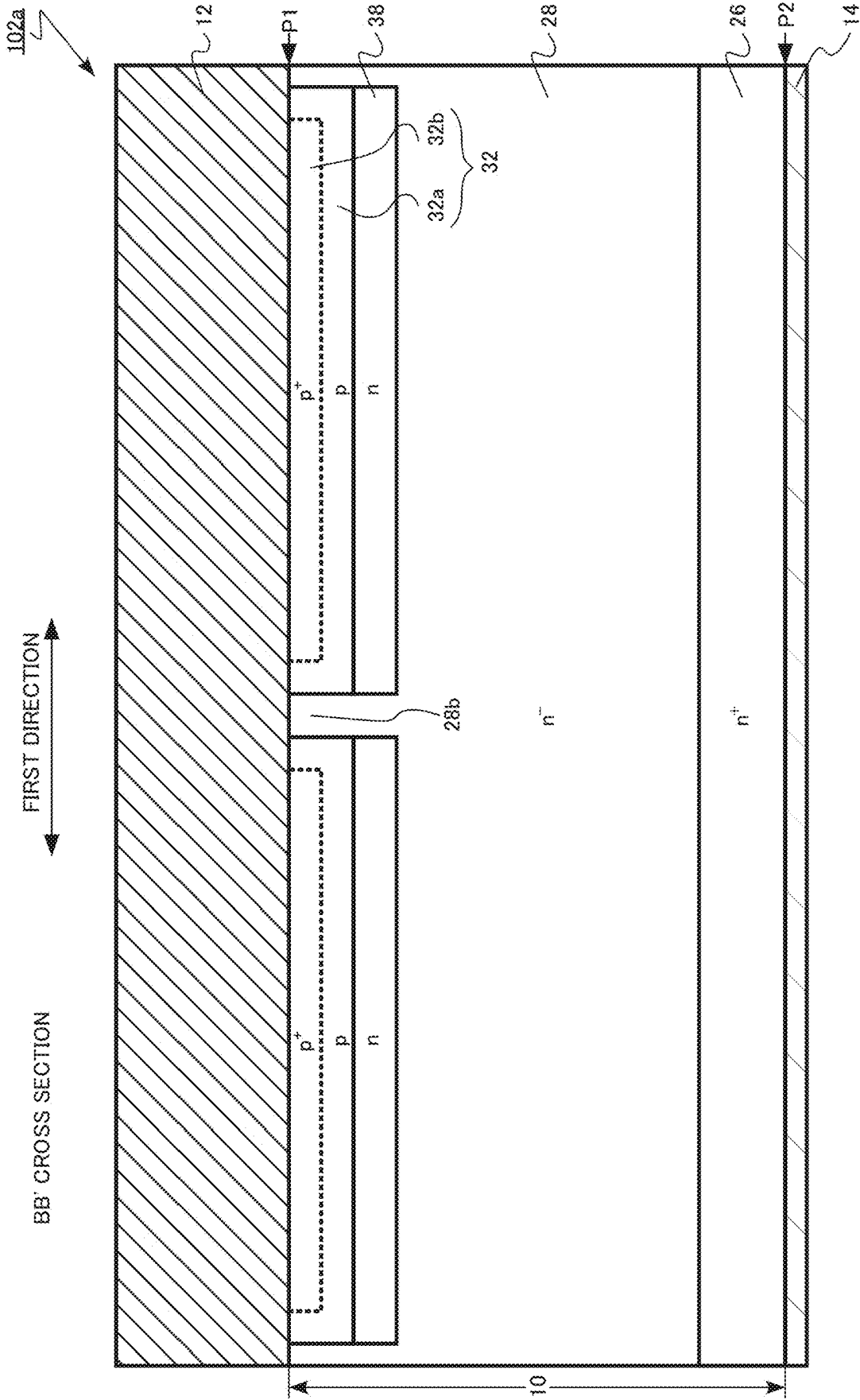


FIG.4



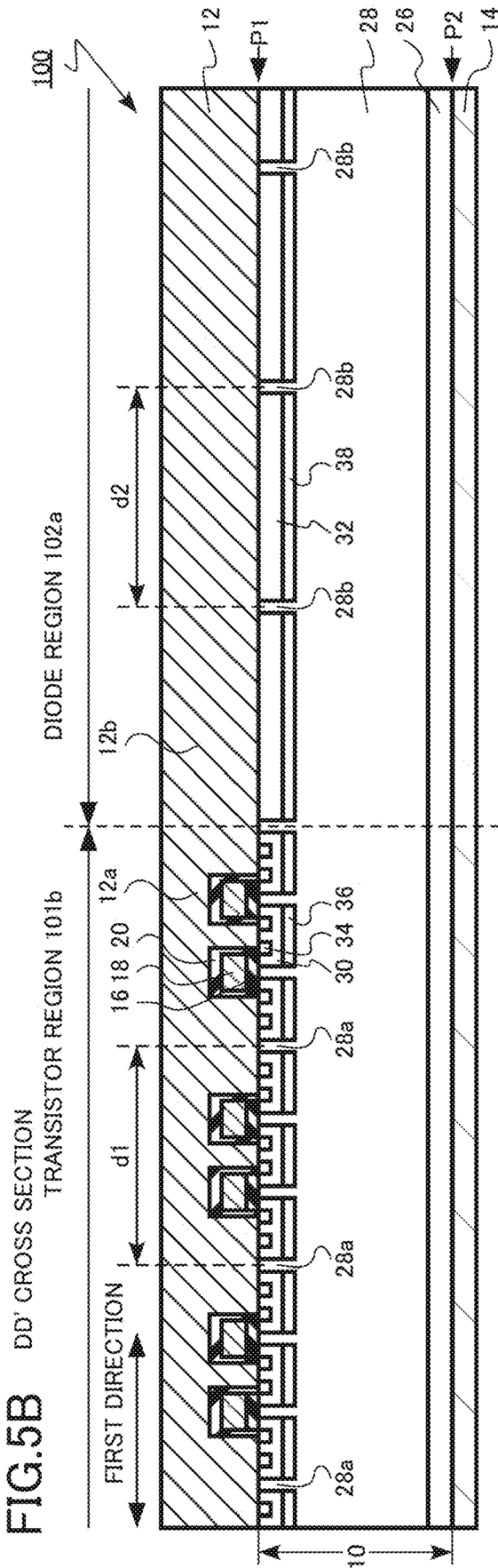
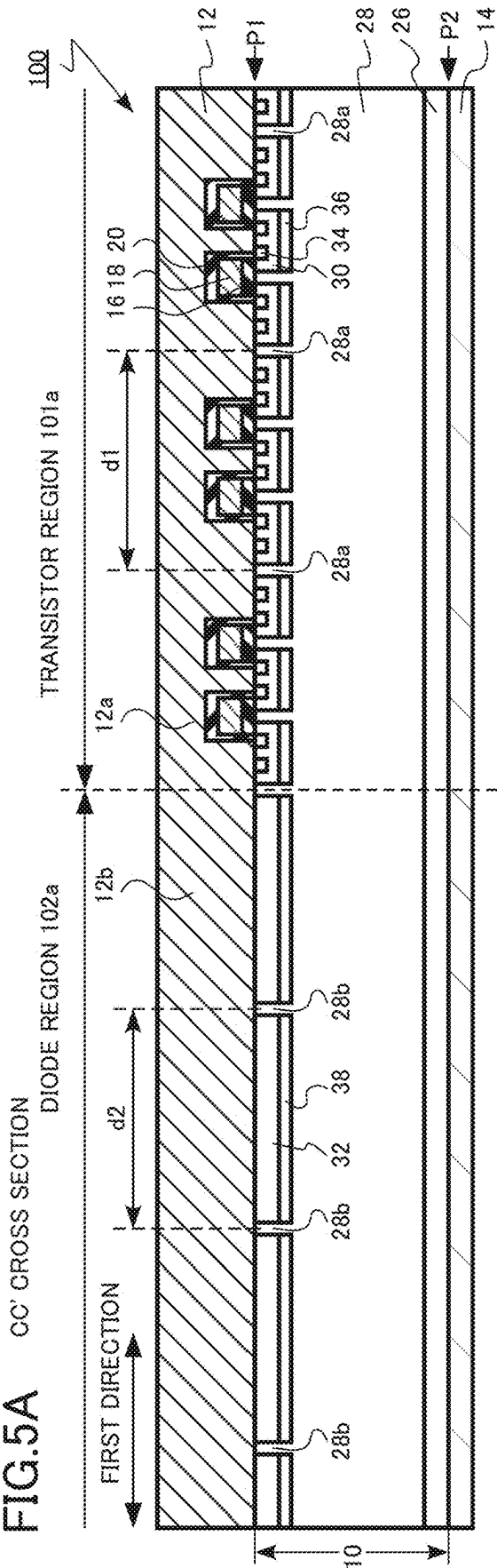


FIG. 6

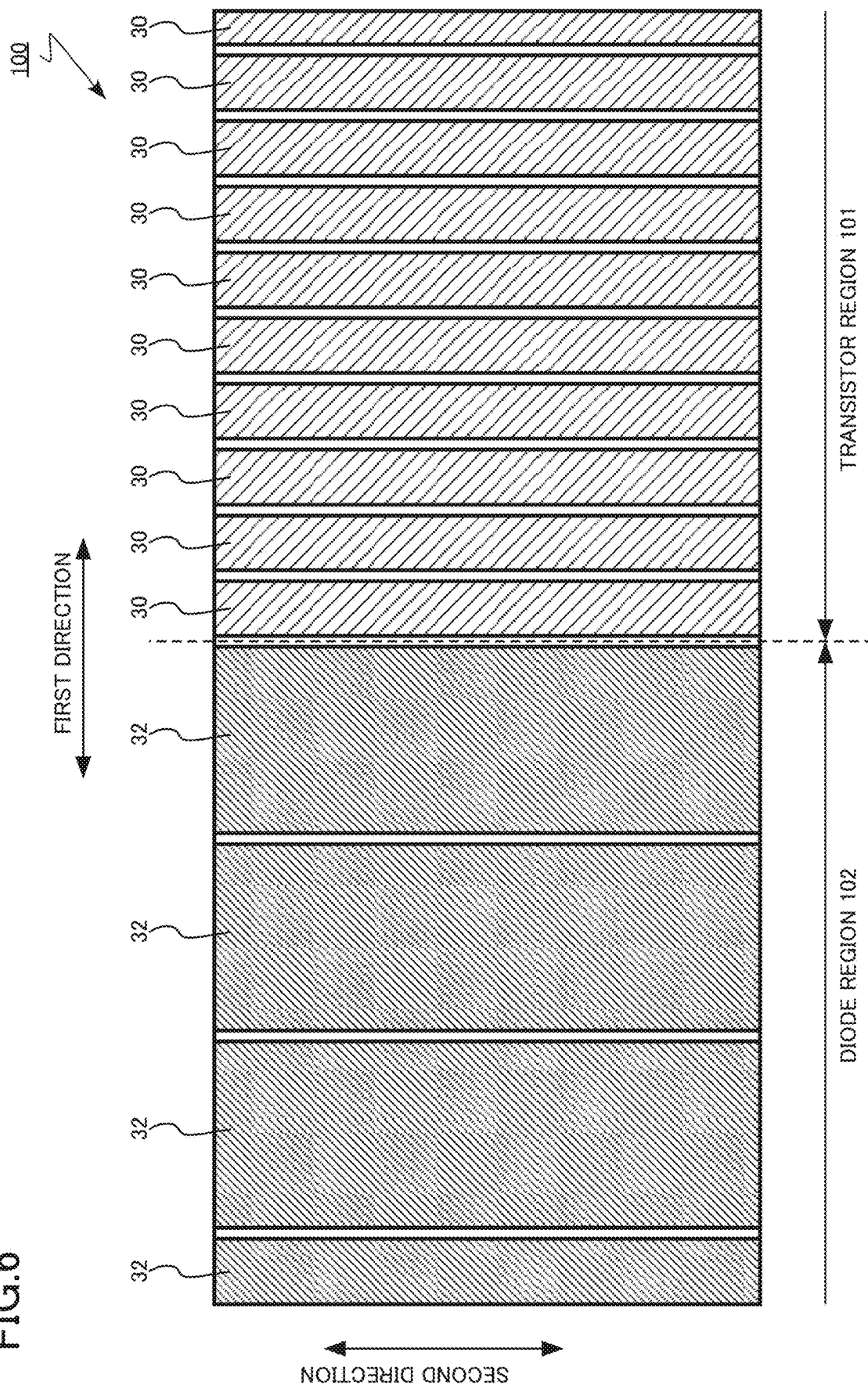


FIG. 7

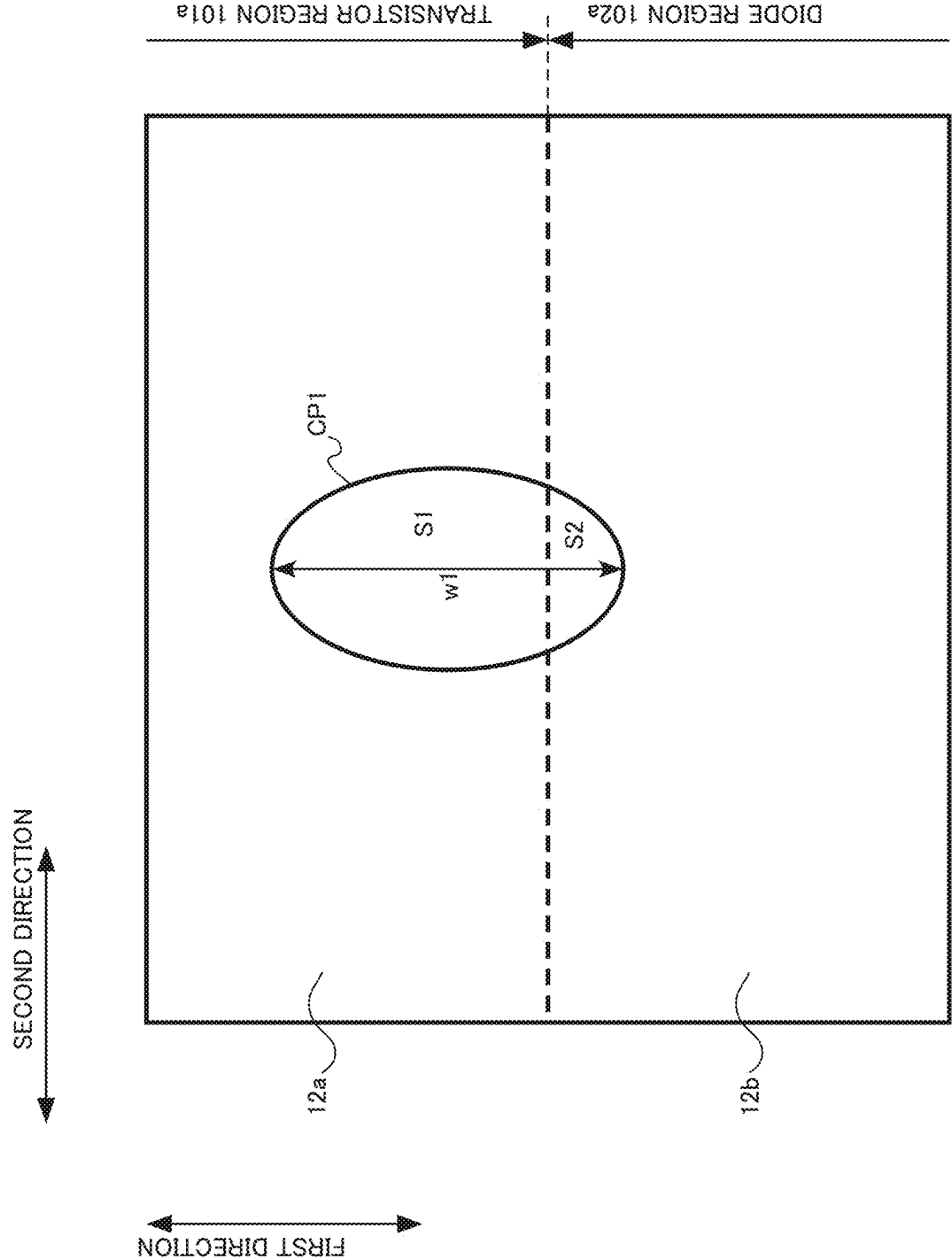


FIG.8A

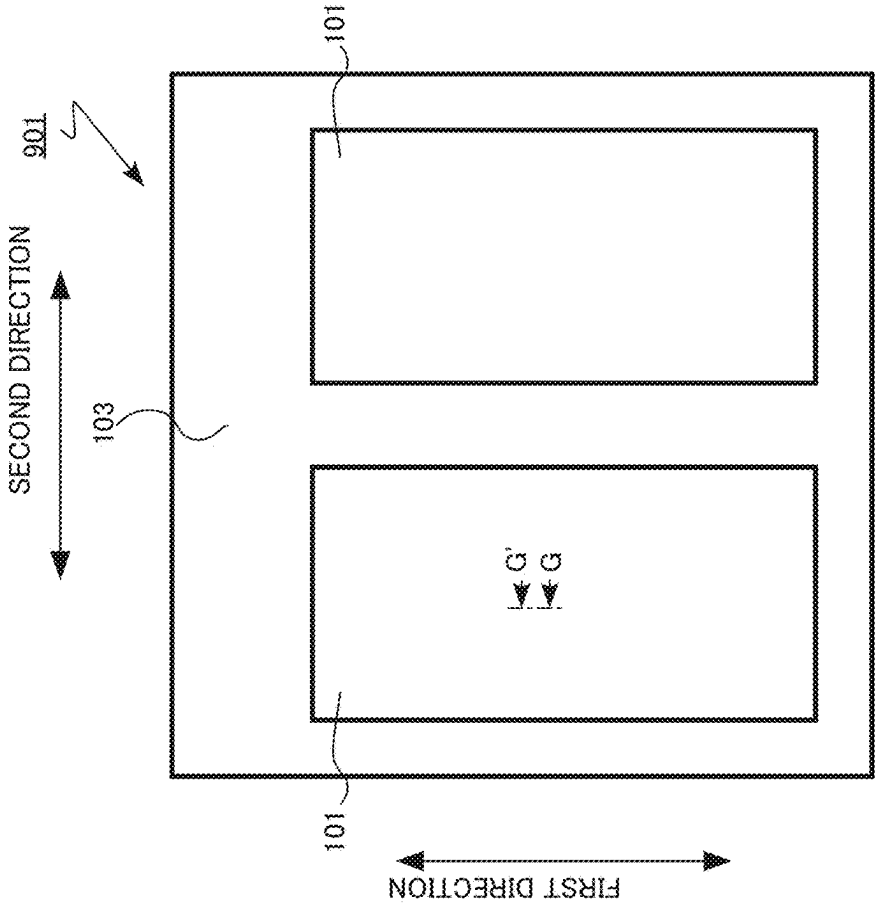


FIG.8B

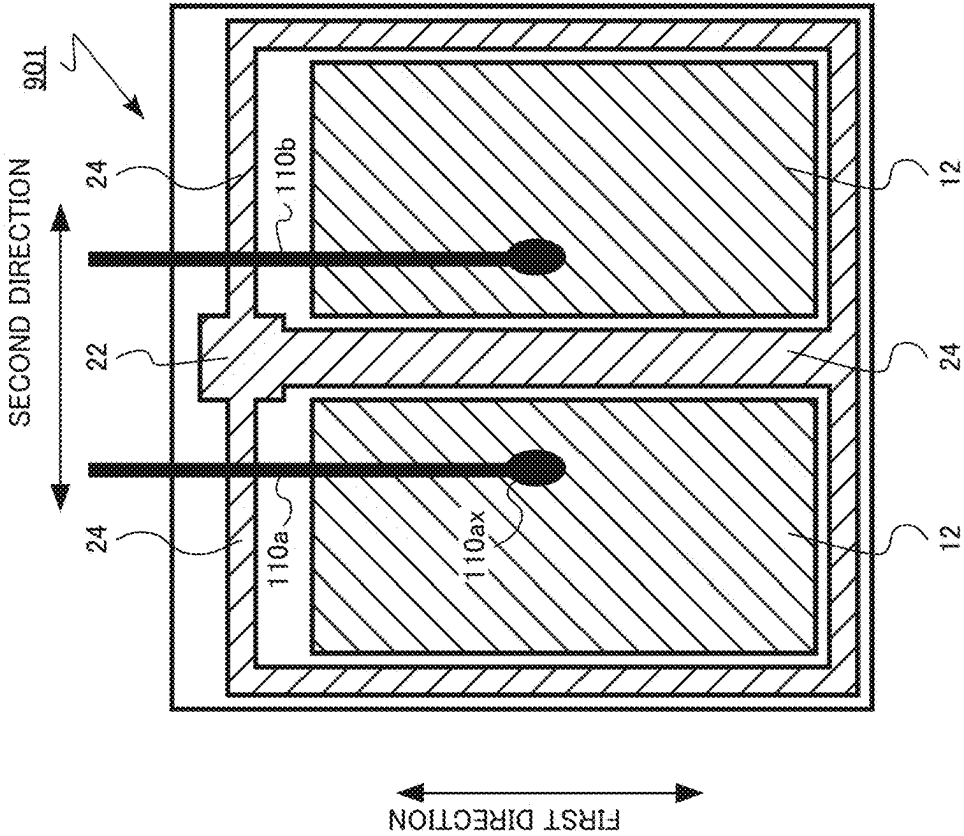


FIG.9

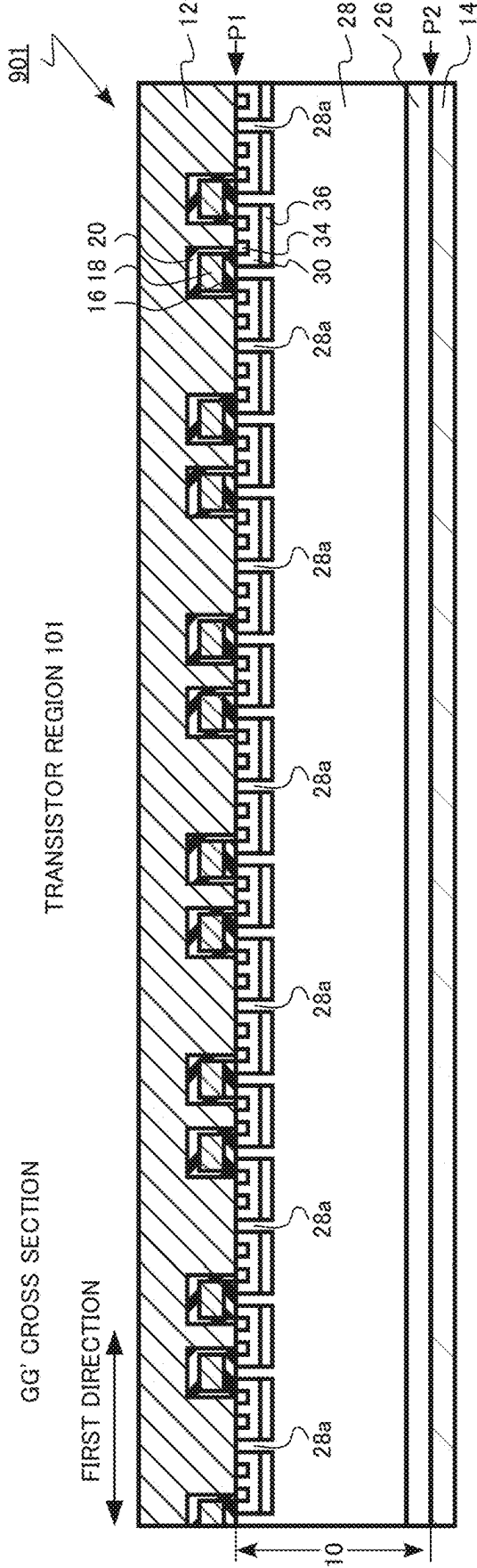


FIG.10

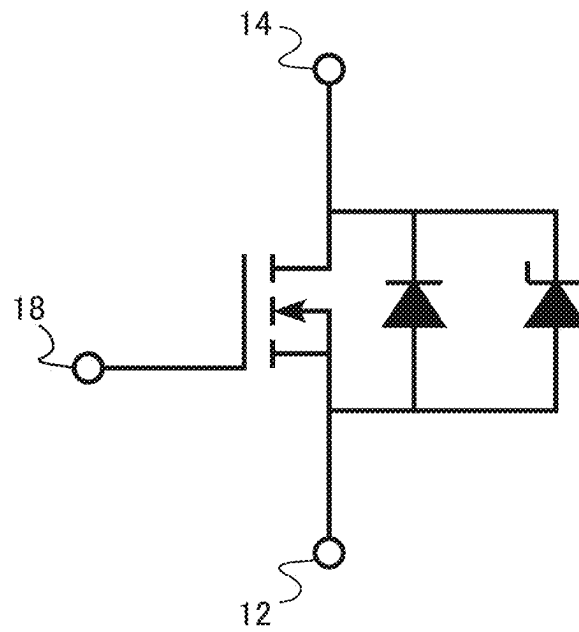


FIG.11A

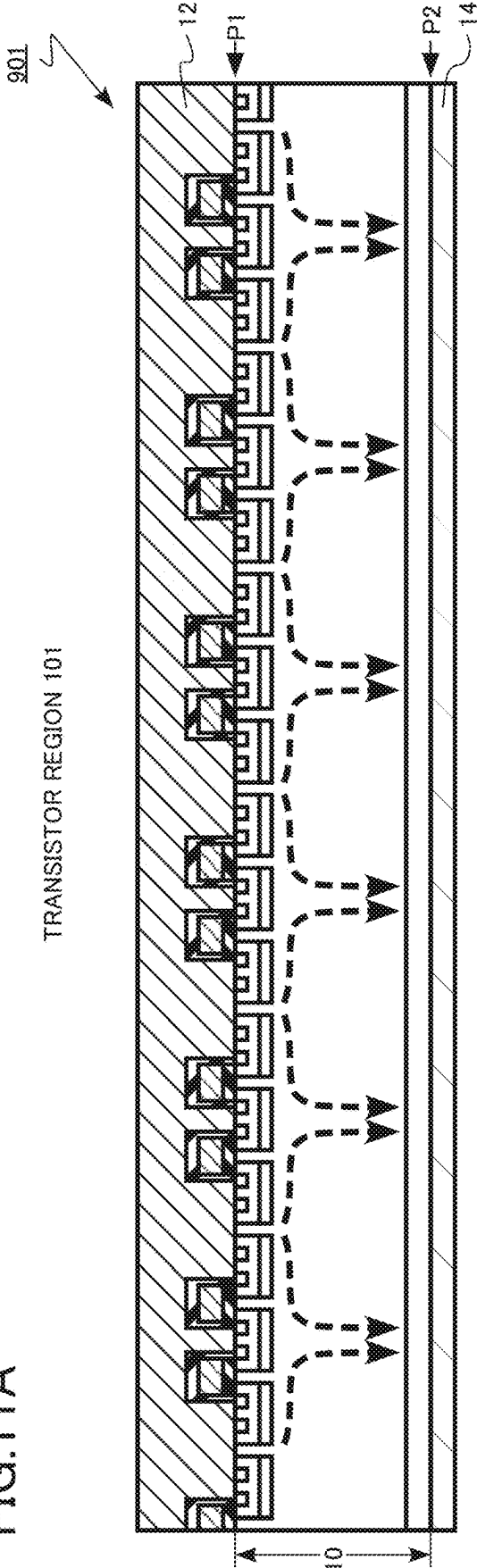


FIG.11B

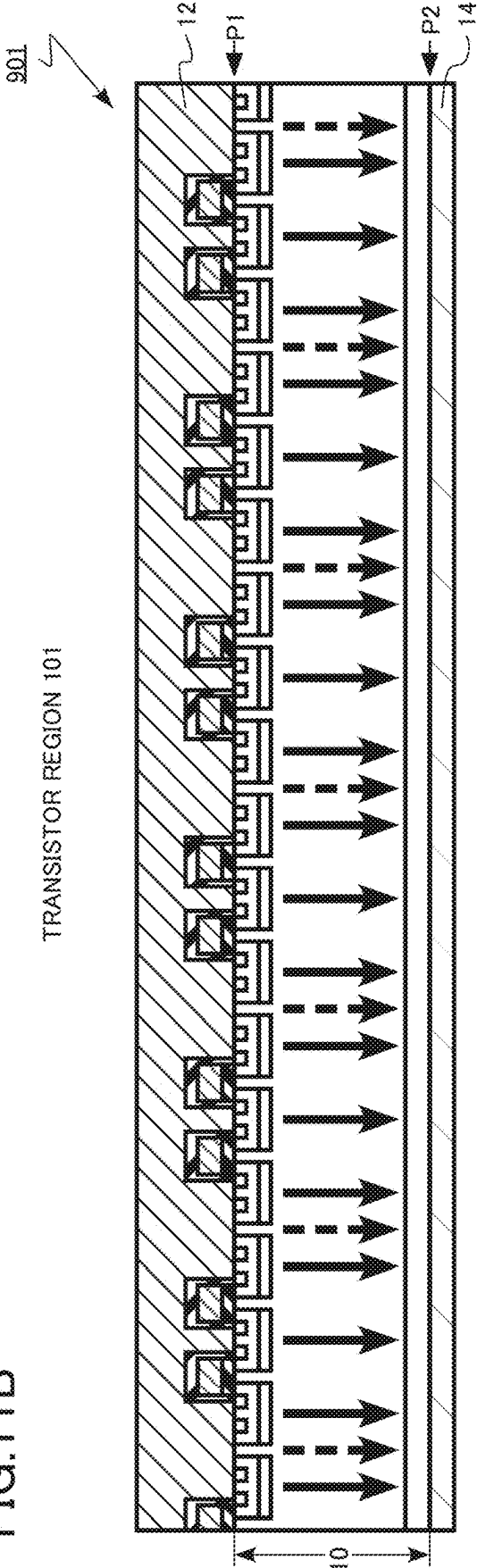


FIG.12

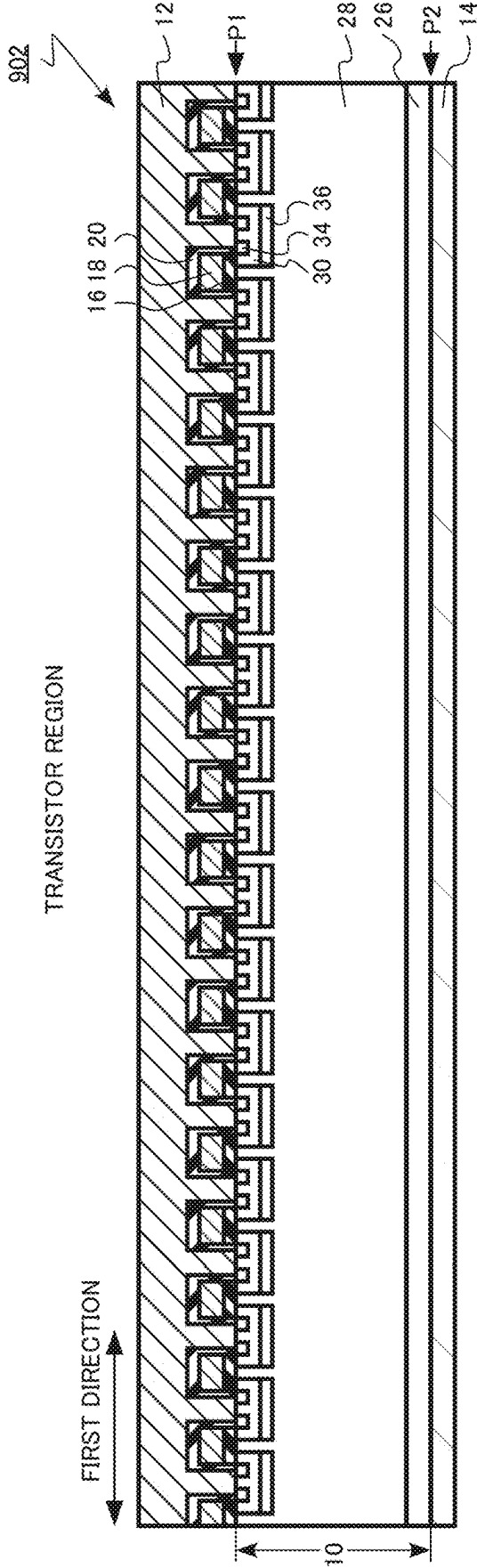


FIG.13

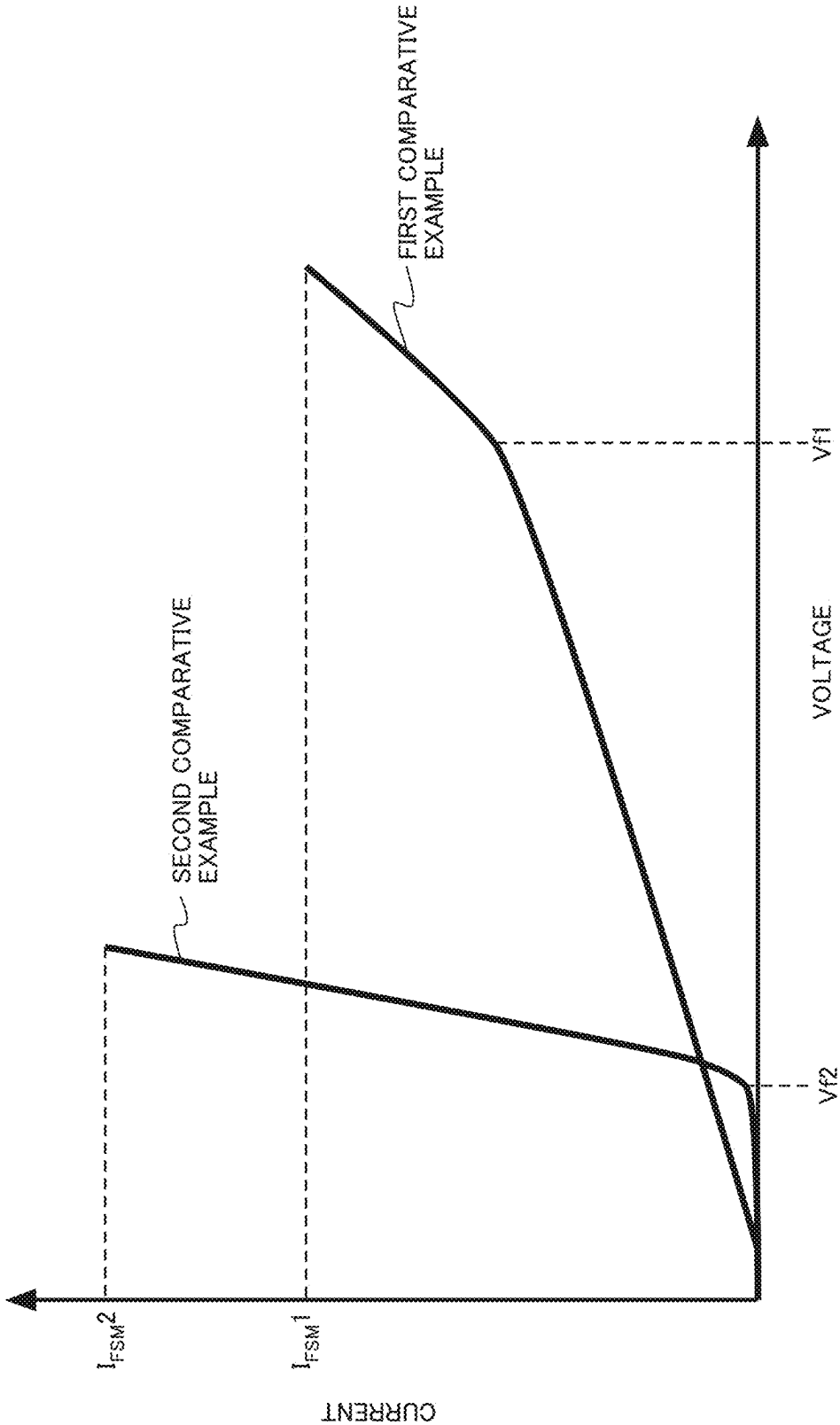


FIG. 14A

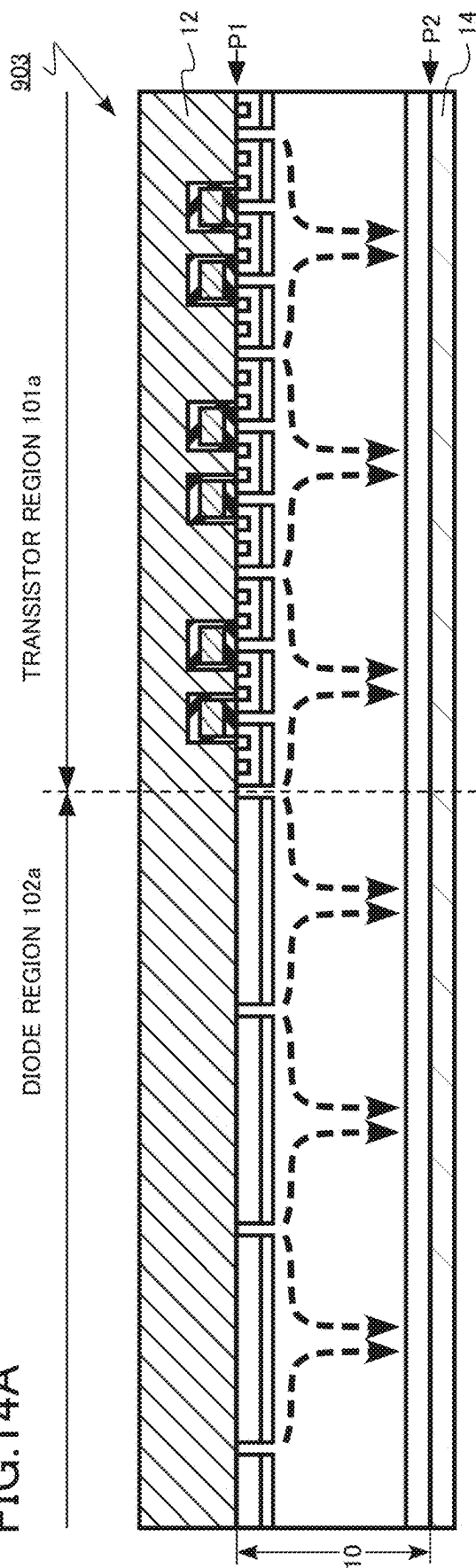


FIG. 14B

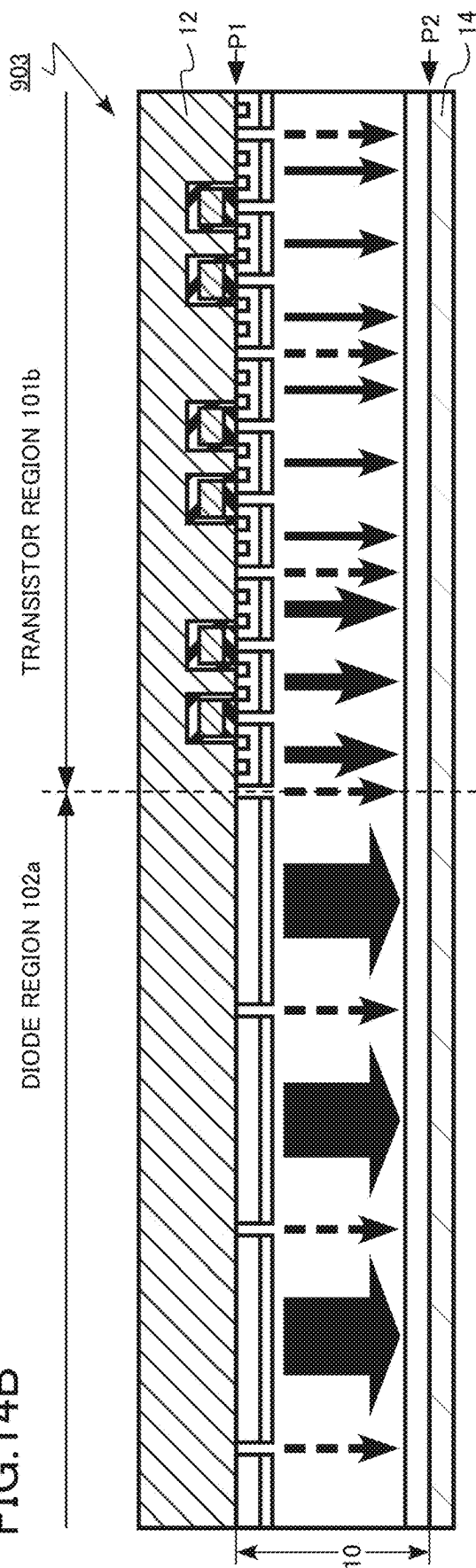


FIG.15

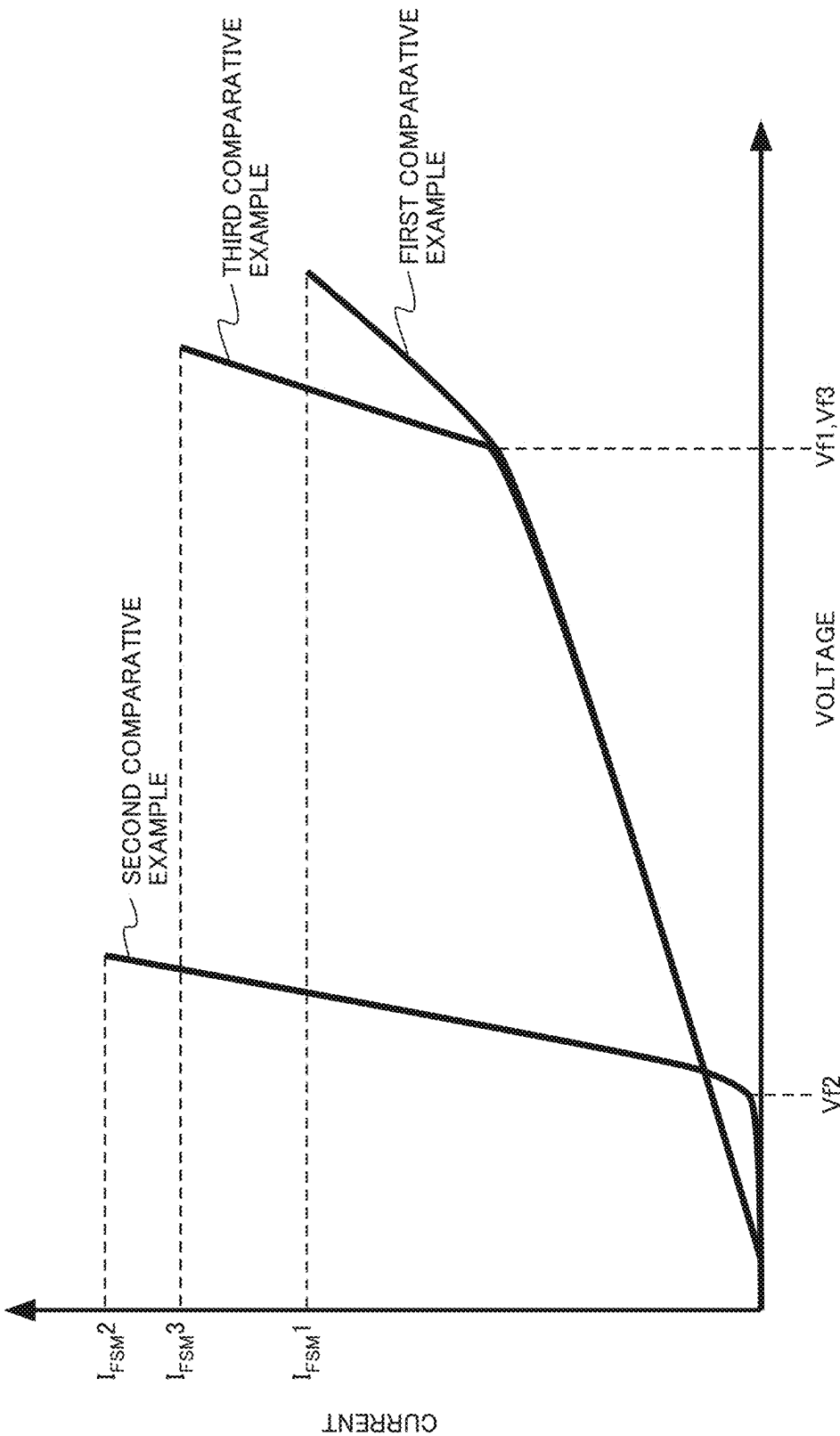


FIG.16A

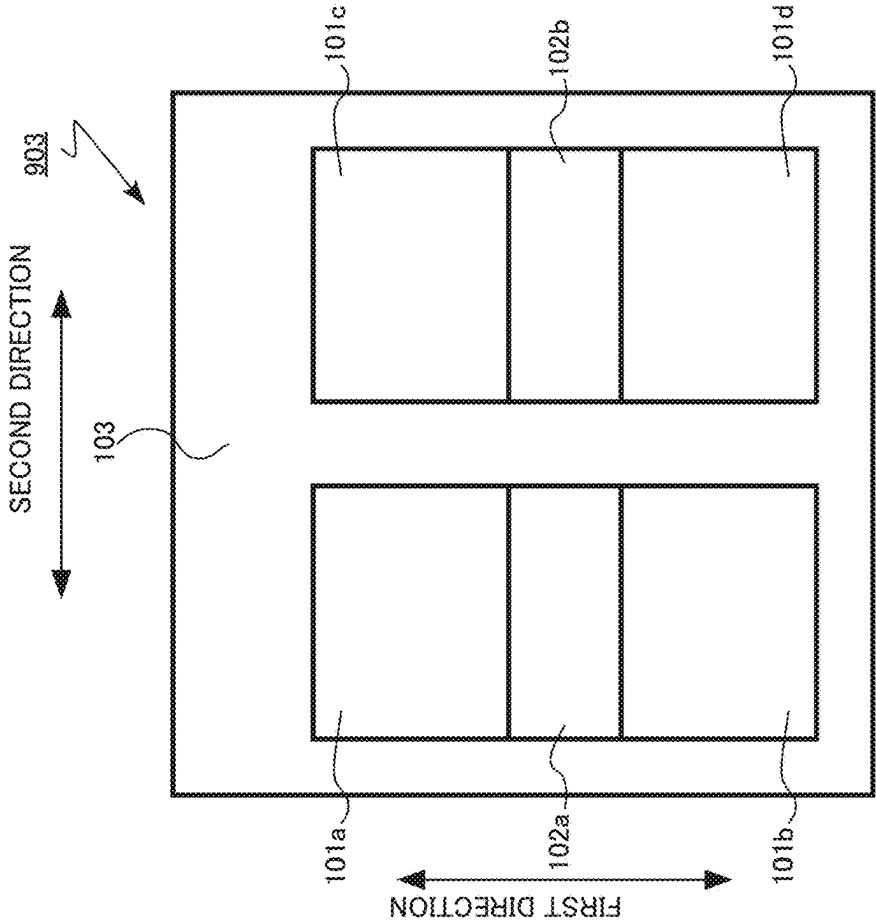


FIG.16B

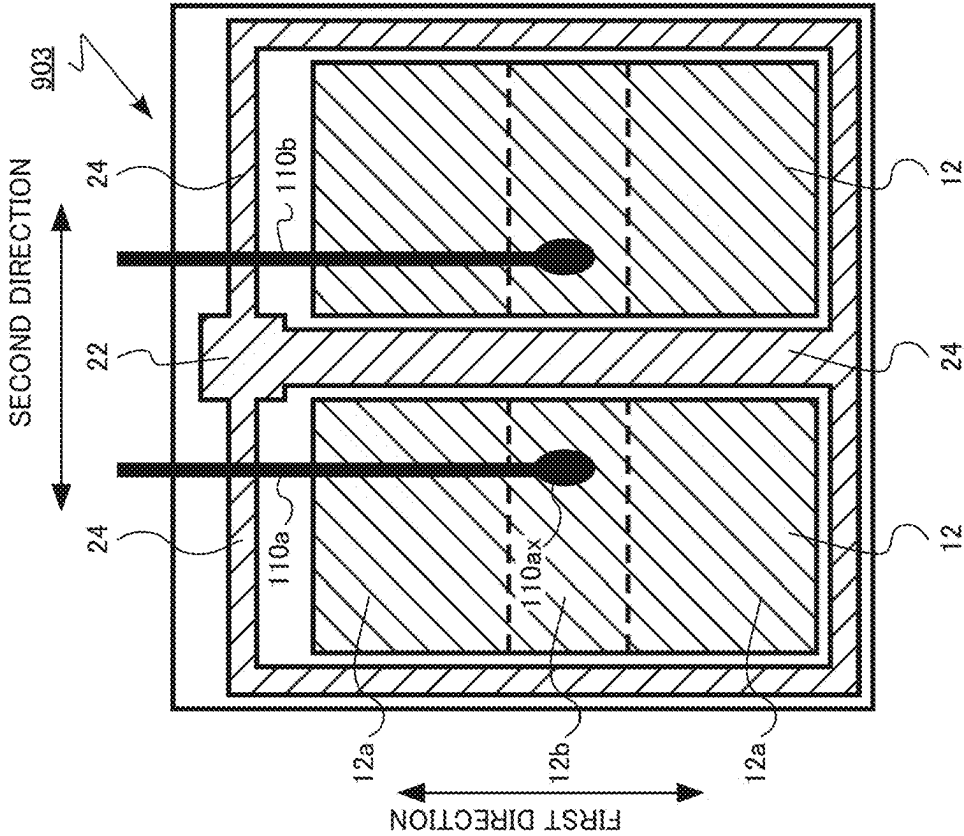


FIG.17

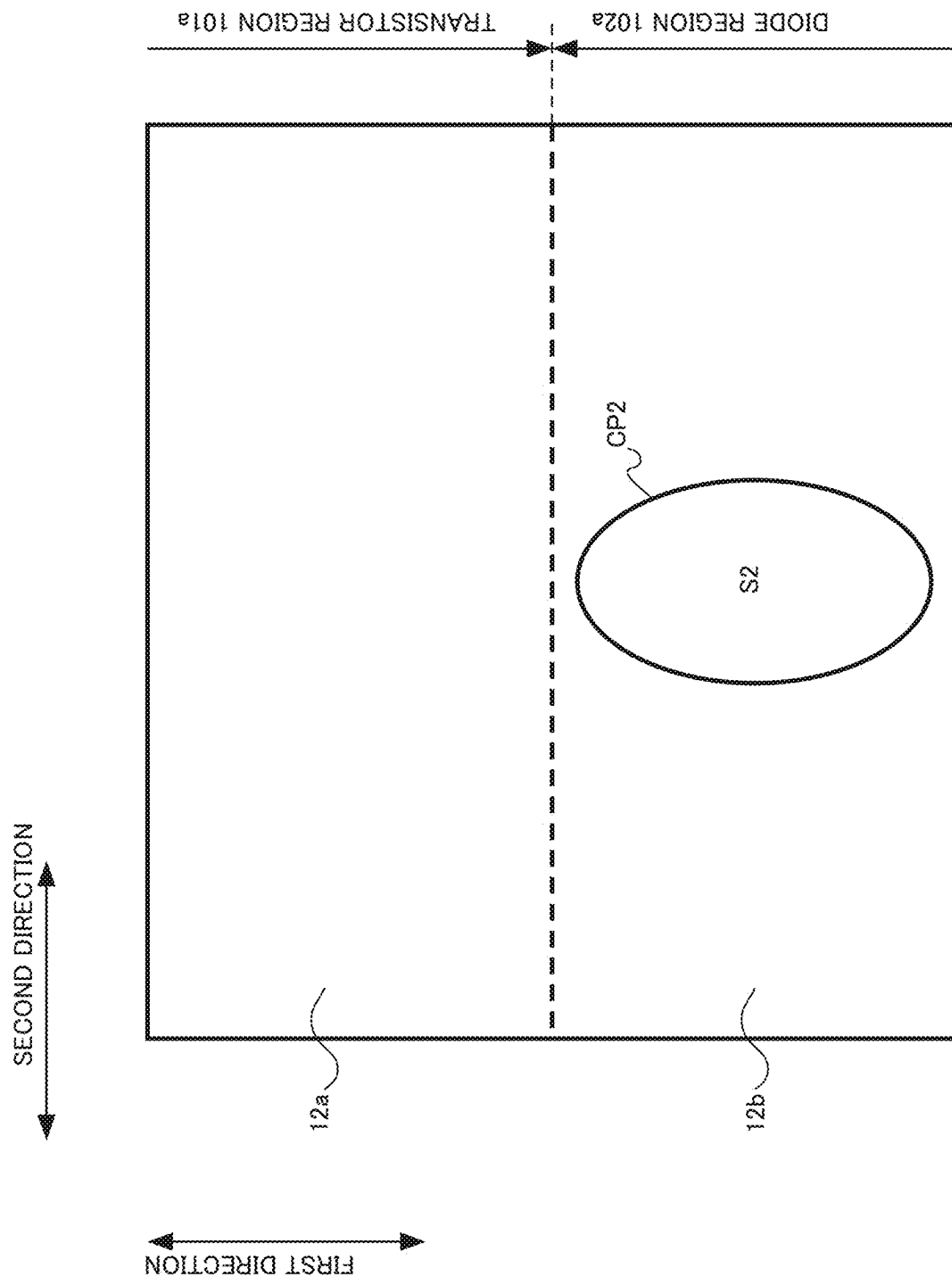


FIG.18A

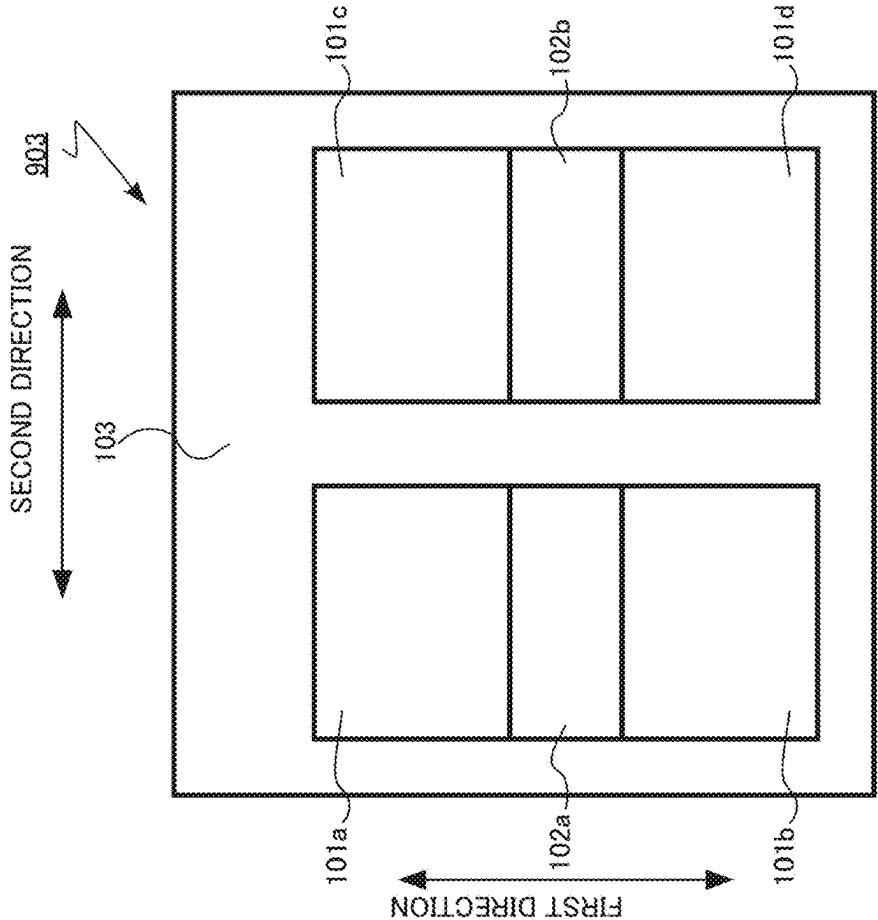


FIG.18B

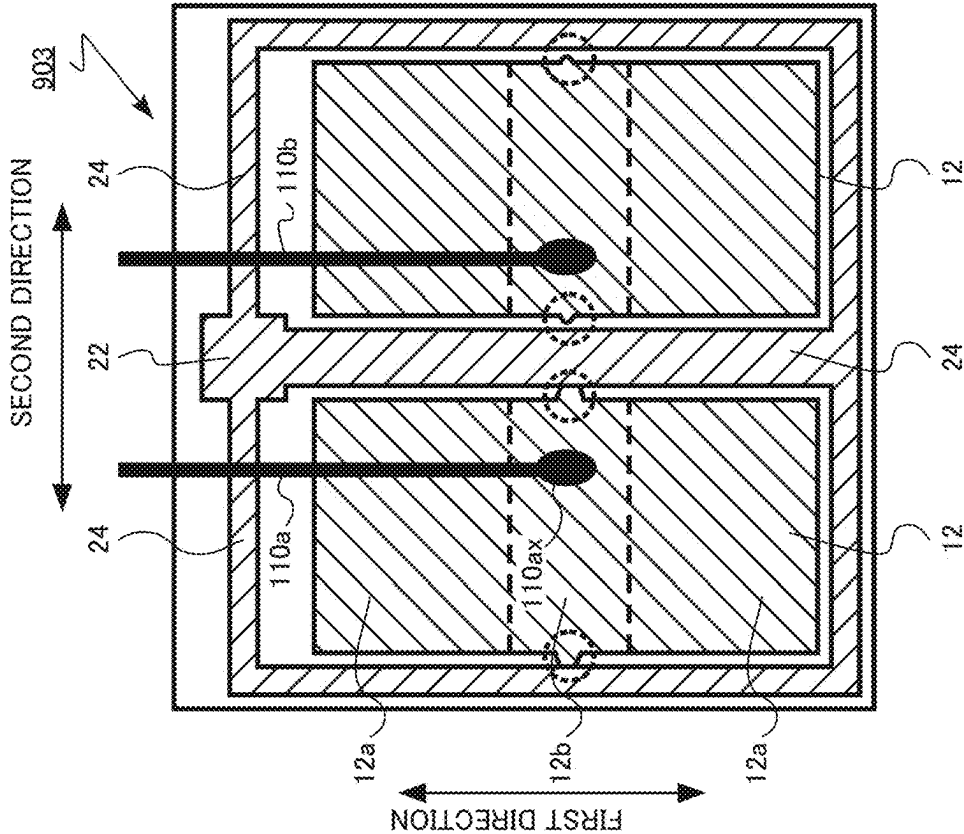


FIG.19A

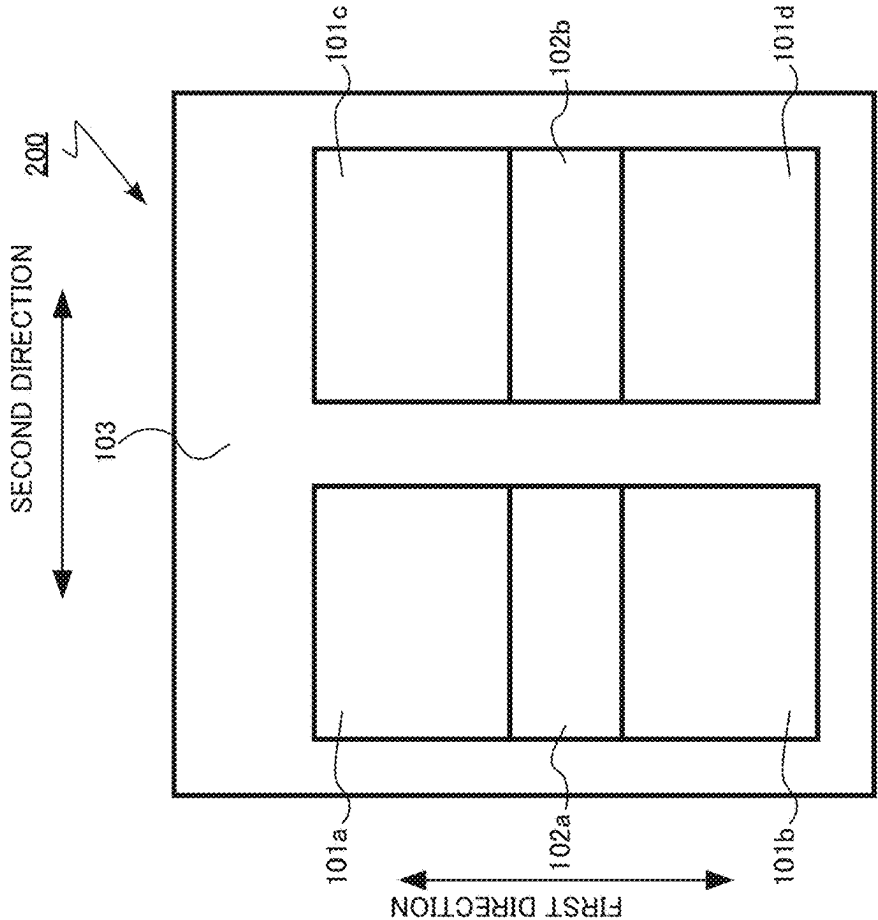


FIG.19B

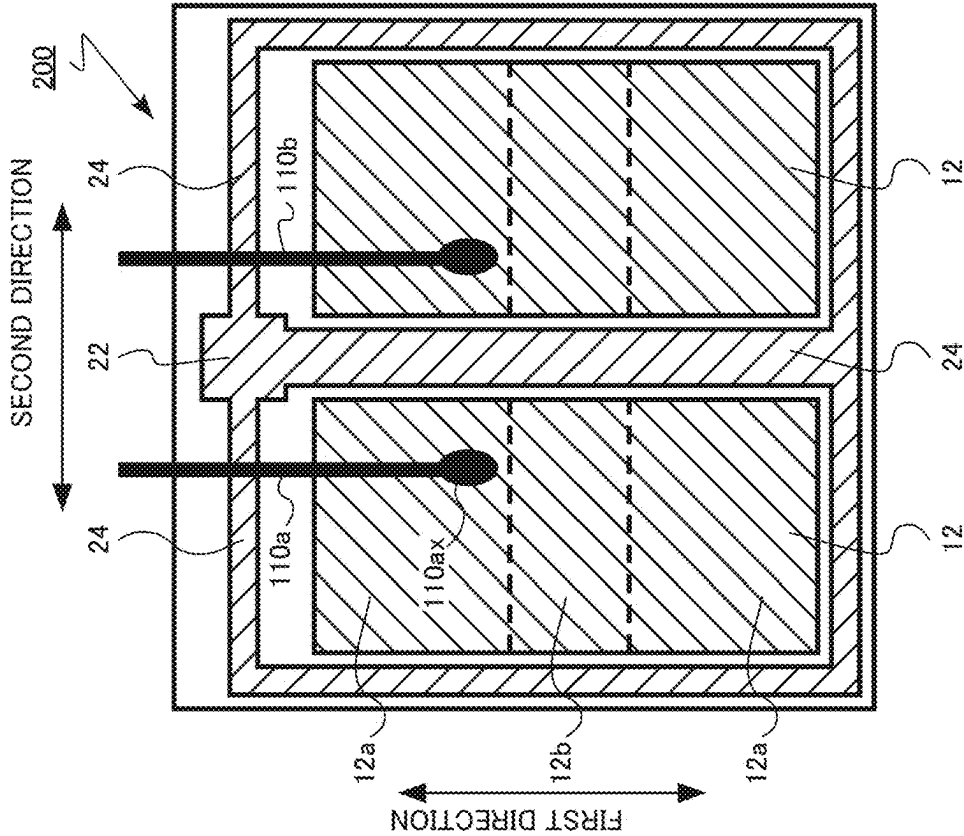


FIG.20

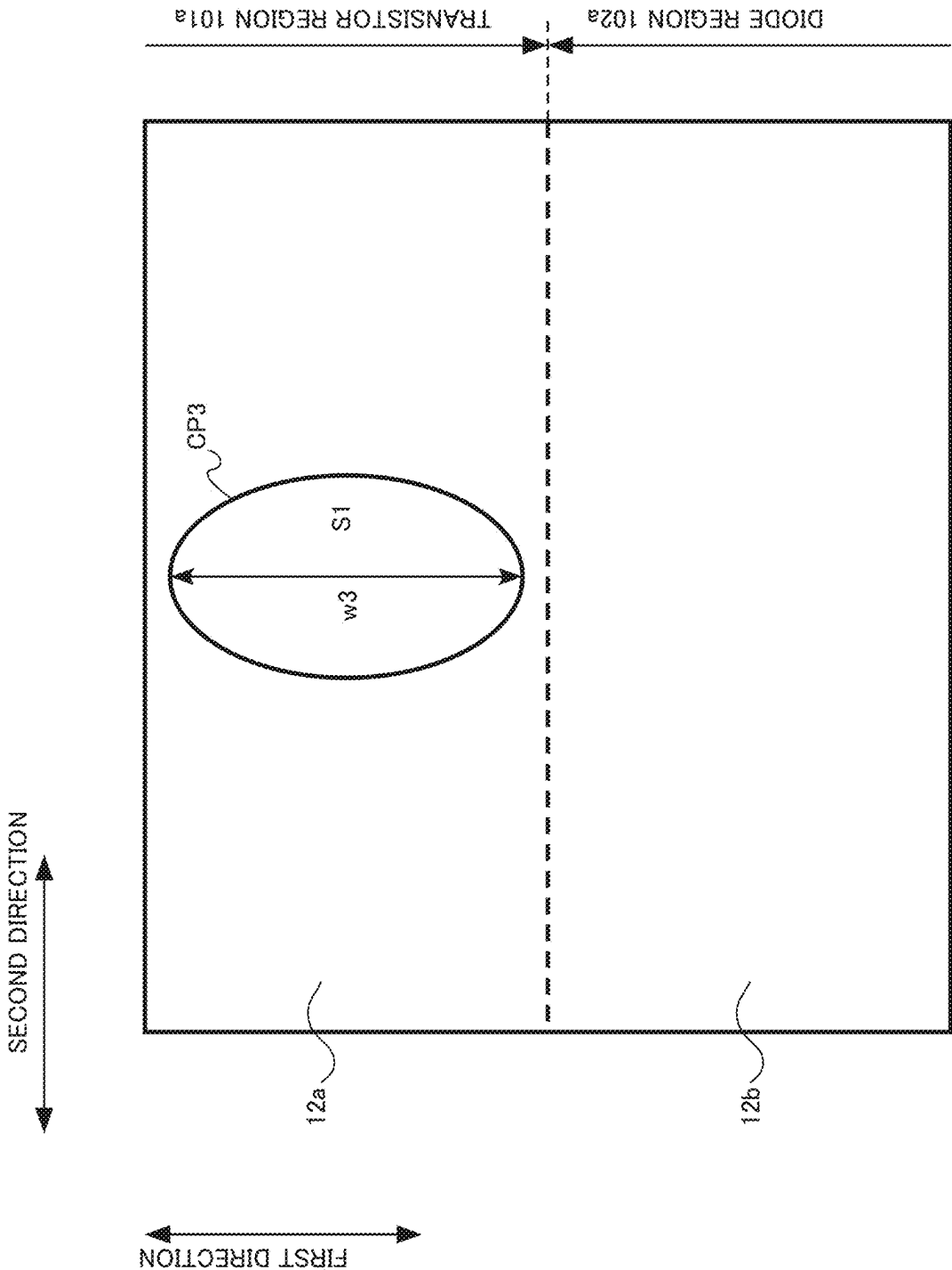


FIG. 21

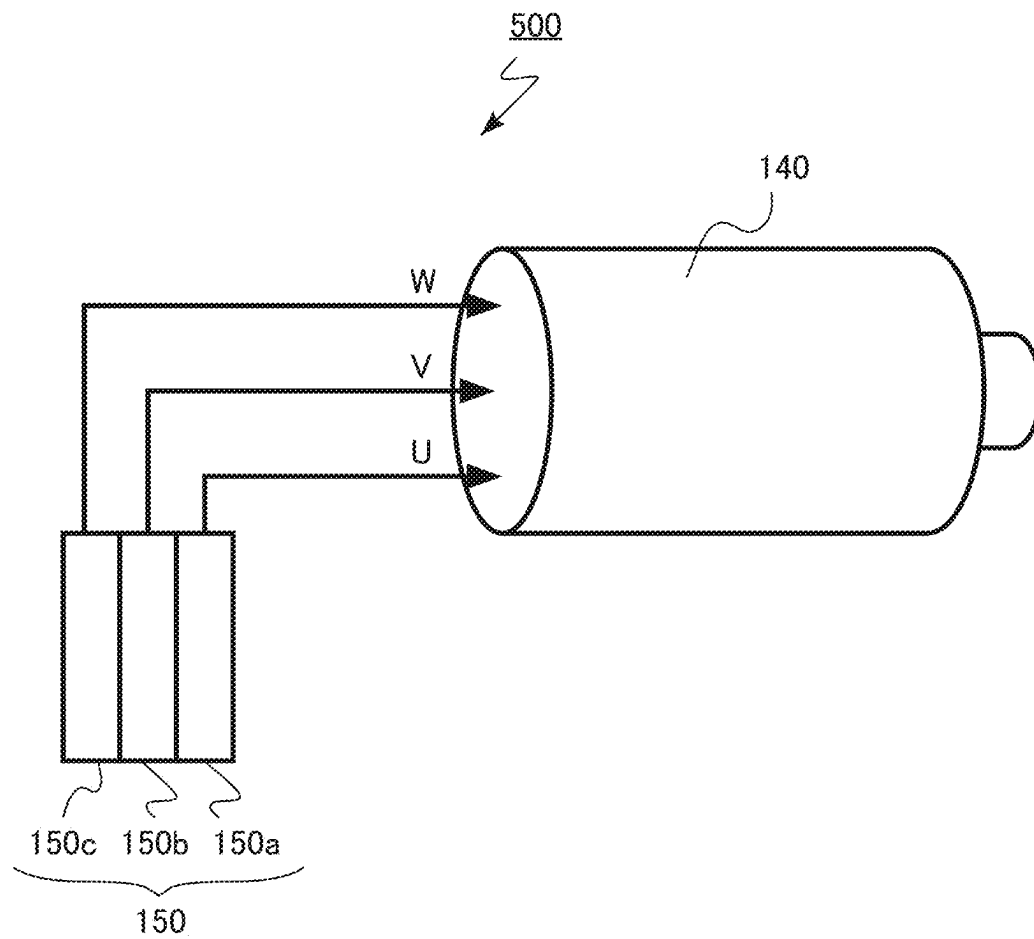


FIG.22

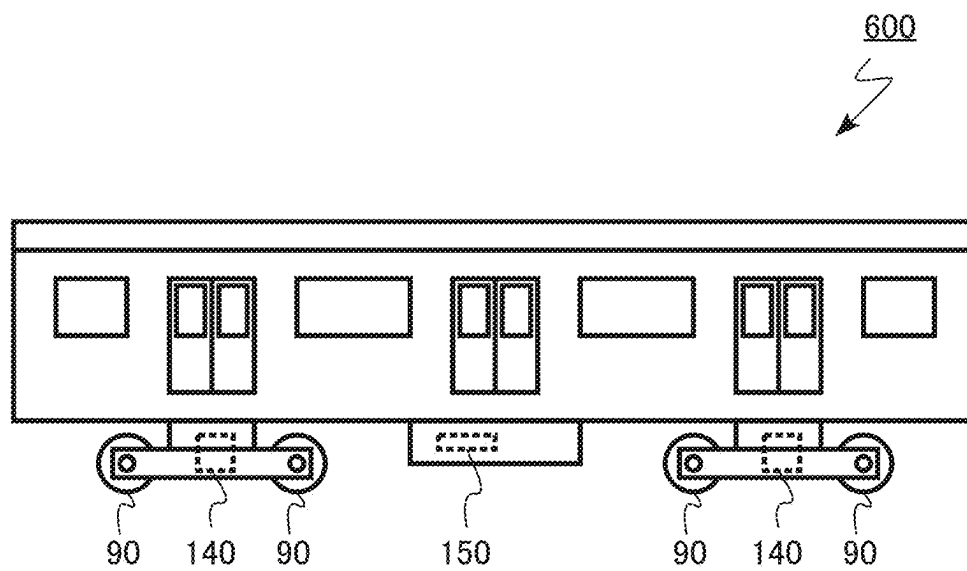


FIG.23

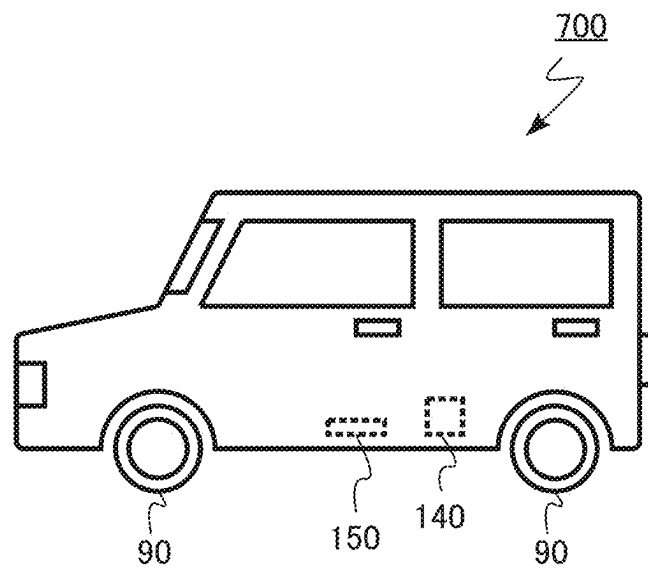
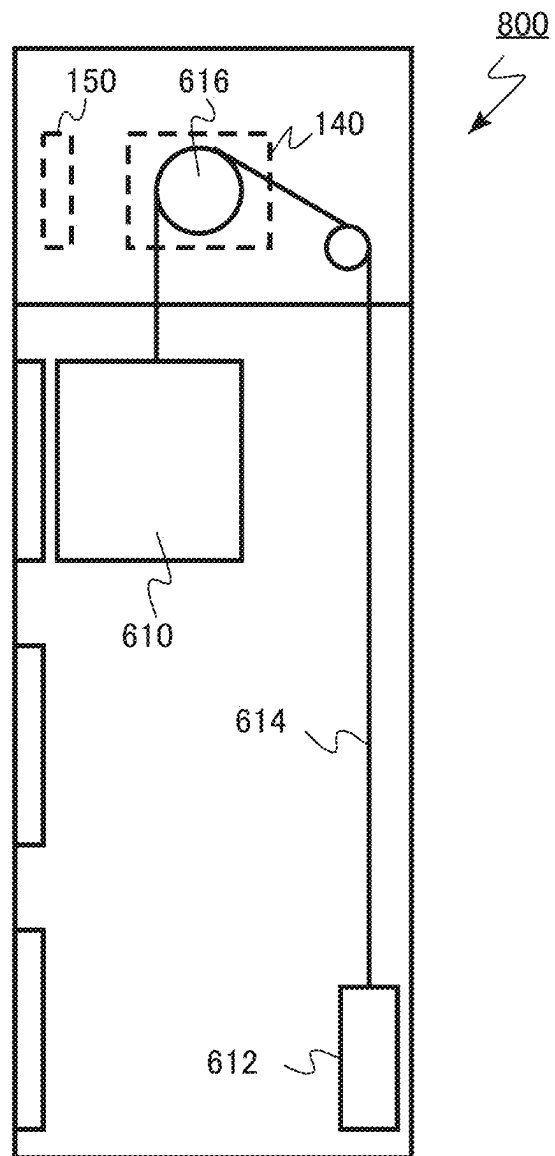


FIG.24



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SEMICONDUCTOR DEVICE, INVERTER CIRCUIT, DRIVING DEVICE, VEHICLE, AND ELEVATOR

CROSS-REFERENCE TO RELATED APPLICATION

This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-139053, filed on Sep. 1, 2022, the entire contents of which are incorporated herein by reference.

FIELD

Embodiments described herein relate generally to a semiconductor device, an inverter circuit, a driving device, a vehicle, and an elevator.

BACKGROUND

Silicon carbide (SiC) is expected as a material for a next-generation semiconductor device. As compared with silicon, silicon carbide has excellent physical properties such as a band gap of three times, a breakdown field strength of about ten times, and a thermal conductivity of about three times. By utilizing this characteristic, for example, a metal oxide semiconductor field effect transistor (MOSFET) capable of operating at a high breakdown voltage, a low loss, and a high temperature can be realized.

A vertical MOSFET using silicon carbide includes a pn junction diode as a built-in diode. For example, the MOSFET is used as a switching element connected to an inductive load. In this case, even when the MOSFET is turned off, it is possible to allow a reflux current to flow using the built-in diode.

However, when a reflux current is caused to flow using a body diode, there is a problem in that a stacking fault grows in a silicon carbide layer due to recombination energy of carriers, and on-resistance of the MOSFET increases. An increase in on-resistance of the MOSFET causes a decrease in reliability of the MOSFET. For example, by providing a Schottky Barrier Diode (SBD) that performs unipolar operation as a built-in diode in the MOSFET, it is possible to suppress growth of the stacking fault in the silicon carbide layer. The reliability of the MOSFET is improved by providing the SBD as a built-in diode in the MOSFET.

A large surge current may flow into the MOSFET instantaneously beyond the steady state. When a large surge current flows, a large surge voltage is applied to generate heat, and the MOSFET is broken. The maximum allowable peak current value (I_{FSM}) of the surge current allowed in the MOSFET is referred to as a surge current tolerance. In the MOSFET having the SBD provided therein, it is desired to improve a surge current tolerance.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A and 1B are schematic diagrams of a semiconductor device according to a first embodiment;

FIGS. 2A and 2B are schematic top views of the semiconductor device according to the first embodiment;

FIG. 3 is a schematic cross-sectional view of the semiconductor device according to the first embodiment;

FIG. 4 is a schematic cross-sectional view of the semiconductor device according to the first embodiment;

FIGS. 5A and 5B are schematic cross-sectional views of the semiconductor device according to the first embodiment;

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FIG. 6 is a schematic top view of the semiconductor device according to the first embodiment;

FIG. 7 is a schematic top view of the semiconductor device according to the first embodiment;

FIGS. 8A and 8B are schematic top views of a semiconductor device according to a first comparative example;

FIG. 9 is a schematic cross-sectional view of the semiconductor device according to the first comparative example;

FIG. 10 is an equivalent circuit diagram of the semiconductor device according to the first comparative example;

FIGS. 11A and 11B are explanatory diagrams of functions and effects of the semiconductor device according to the first embodiment;

FIG. 12 is a schematic cross-sectional view of a semiconductor device according to a second comparative example;

FIG. 13 is an explanatory diagram of functions and effects of the semiconductor device according to the first embodiment;

FIGS. 14A and 14B are explanatory diagrams of functions and effects of the semiconductor device according to the first embodiment;

FIG. 15 is an explanatory diagram of functions and effects of the semiconductor device according to the first embodiment;

FIGS. 16A and 16B are schematic top views of a semiconductor device according to a third comparative example;

FIG. 17 is a schematic top view of a semiconductor device according to a third comparative example;

FIGS. 18A and 18B are explanatory diagrams of a problem of the semiconductor device according to the third comparative example;

FIGS. 19A and 19B are schematic top views of a semiconductor device according to a second embodiment;

FIG. 20 is a schematic top view of the semiconductor device according to the second embodiment;

FIG. 21 is a schematic diagram of a driving device according to a third embodiment;

FIG. 22 is a schematic diagram of a vehicle according to a fourth embodiment;

FIG. 23 is a schematic diagram of a vehicle according to a fifth embodiment; and

FIG. 24 is an elevator according to a sixth embodiment.

DETAILED DESCRIPTION

A semiconductor device according to an embodiment includes: a semiconductor chip including a plurality of transistor regions and at least one diode region, wherein the transistor regions include a silicon carbide layer having a first plane and a second plane facing the first plane, the silicon carbide layer including a first silicon carbide region of n-type having a plurality of first portions in contact with the first plane, a second silicon carbide region of p-type provided between the first silicon carbide region and the first plane, and a third silicon carbide region of n-type provided between the second silicon carbide region and the first plane, a first electrode in contact with the first portions, the second silicon carbide region, and the third silicon carbide region, a second electrode in contact with the second plane, a gate electrode facing the second silicon carbide region, and a gate insulating layer provided between the gate electrode and the second silicon carbide region, wherein the at least one diode region includes the silicon carbide layer including the first silicon carbide region of n-type having a plurality of second portions in contact with the first plane and a fourth silicon

carbide region of p-type provided between the first silicon carbide region and the first plane, the first electrode in contact with the second portions and the fourth silicon carbide region, and the second electrode, wherein an occupied area per unit area of the fourth silicon carbide region projected onto the first plane is larger than an occupied area per the unit area of the second silicon carbide region projected onto the first plane, and wherein a first diode region, which is one of the at least one diode region, is provided between a first transistor region, which is one of the transistor regions, and a second transistor region, which is one of the transistor regions provided in a first direction with respect to the first transistor region; and a conductor having one end in contact with the first electrode and applying a voltage to the first electrode, wherein the first electrode includes a first region in the transistor regions and a second region in the at least one diode region, and wherein a first contact area between the conductor and the first region is larger than a second contact area between the conductor and the second region.

Hereinafter, embodiments of the present disclosure will be described with reference to the drawings. In the following description, the same or similar members and the like are denoted by the same reference numerals, and the description of the members and the like once described may be appropriately omitted.

In the following description, when notations of n^+ , n , and n^- and p^+ , p , and p^- are used, the above notations represent relative levels of impurity concentration in each conductive type. That is, n^+ indicates that the n-type impurity concentration is relatively higher than n , and n^- indicates that the n-type impurity concentration is relatively lower than n . In addition, p^+ indicates that the p-type impurity concentration is relatively higher than p , and p^- indicates that the p-type impurity concentration is relatively lower than p . In addition, an n^+ type and an n^- type may be simply referred to as an n type, and a p^+ type and a p^- type may be simply referred to as a p type.

The impurity concentration can be measured by, for example, secondary ion mass spectrometry (SIMS). Further, the relative level of the impurity concentration can also be determined from the level of carrier concentration obtained by, for example, scanning capacitance microscopy (SCM). In addition, distances such as a depth and a thickness of an impurity region can be obtained by, for example, SIMS. Furthermore, distances such as a depth, a thickness, a width, and an interval of the impurity region can be obtained from, for example, a composite image of an SCM image and an atomic force microscope (AFM) image.

In the present specification, impurity concentration of a semiconductor region means the maximum impurity concentration of the semiconductor region unless otherwise stated.

First Embodiment

A semiconductor device according to a first embodiment includes a semiconductor chip including a plurality of transistor regions and at least one diode region. The plurality of transistor regions include a silicon carbide layer having a first plane and a second plane facing the first plane, the silicon carbide layer including a first silicon carbide region of n-type having a plurality of first portions in contact with the first plane, a second silicon carbide region of p-type provided between the first silicon carbide region and the first plane, and a third silicon carbide region of n-type provided between the second silicon carbide region and the first plane,

a first electrode in contact with the plurality of first portions, the second silicon carbide region, and the third silicon carbide region, a second electrode in contact with the second plane, a gate electrode facing the second silicon carbide region, and a gate insulating layer provided between the gate electrode and the second silicon carbide region. The at least one diode region includes the silicon carbide layer including the first silicon carbide region of n-type having a plurality of second portions in contact with the first plane, and a fourth silicon carbide region of p-type provided between the first silicon carbide region and the first plane, the first electrode in contact with the plurality of second portions and the fourth silicon carbide region, and the second electrode. An occupied area per unit area of the fourth silicon carbide region projected onto the first plane is larger than an occupied area per unit area of the second silicon carbide region projected onto the first plane. A first diode region, which is one of the at least one diode region, is provided between a first transistor region, which is one of the plurality of transistor regions, and a second transistor region, which is one of the plurality of transistor regions provided in a first direction with respect to the first transistor region. The semiconductor device according to the first embodiment includes a conductor having one end in contact with the first electrode and applying a voltage to the first electrode. The first electrode includes a first region in the plurality of transistor regions and a second region in the at least one diode region. A first contact area between the conductor and the first region is larger than a second contact area between the conductor and the second region.

The semiconductor device according to the first embodiment is a discrete device **1000**. In the case of the discrete device **1000**, the discrete device **1000** has one MOSFET **100** mounted thereon with a sealing resin. The MOSFET **100** is an example of a semiconductor chip.

The MOSFET **100** included in the discrete device **1000** is a vertical MOSFET of a planar gate type using silicon carbide. The MOSFET **100** is, for example, a double implantation MOSFET (DIMOSFET) in which a body region and a source region are formed by ion implantation. The MOSFET **100** is a MOSFET including a Schottky barrier diode (SBD) as a built-in diode. The MOSFET **100** is a vertical MOSFET of an n-channel type using electrons as carriers.

FIGS. 1A and 1B are schematic diagrams of the semiconductor device according to the first embodiment. FIG. 1A is a top view of the discrete device **1000**. FIG. 1B is a cross-sectional view of the discrete device **1000**, taken along line K-K' in FIG. 1A.

The discrete device **1000** includes the MOSFET **100**, a first bonding wire **110a** (conductor), a second bonding wire **110b**, a third bonding wire **110c**, a metal bed **120** (first metal layer), a first metal lead **130a** (second metal layer), a second metal lead **130b**, a third metal lead **130c**, and a sealing resin **145**.

The MOSFET **100** includes a silicon carbide layer **10**, a source electrode **12** (first electrode), a drain electrode **14** (second electrode), a gate electrode pad **22**, and a gate wiring **24**.

The MOSFET **100** is disposed on the metal bed **120**. The MOSFET **100** is connected to the metal bed **120**.

The metal bed **120** faces the drain electrode **14** of the MOSFET **100**. The metal bed **120** is an example of a first metal layer. The metal bed **120** is in contact with the drain electrode **14**. The metal bed **120** is electrically connected to the drain electrode **14**.

The metal bed **120** is made of metal. The metal bed **120** is, for example, a copper-based alloy or an iron-nickel alloy.

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The first metal lead **130a** extends in the first direction. The first metal lead **130a** is an example of a second metal layer.

The first metal lead **130a** is an electrode terminal. The first metal lead **130a** has a function of applying a voltage to the semiconductor chip **100** from the outside of the discrete device **1000**.

The first metal lead **130a** is metal. The first metal lead **130a** is, for example, a copper-based alloy or an iron-nickel alloy.

The second metal lead **130b** extends in the first direction. The second metal lead **130b** is an electrode terminal. The second metal lead **130b** has a function of applying a voltage to the semiconductor chip **100** from the outside of the discrete device **1000**.

The second metal lead **130b** is metal. The second metal lead **130b** is, for example, a copper-based alloy or an iron-nickel alloy.

The third metal lead **130c** extends in the first direction. The third metal lead **130c** is an electrode terminal. The third metal lead **130c** has a function of applying a voltage to the semiconductor chip **100** from the outside of the discrete device **1000**. The third metal lead **130c** is connected to the metal bed **120**.

The third metal lead **130c** is metal. The third metal lead **130c** is, for example, a copper-based alloy or an iron-nickel alloy.

The first bonding wire **110a** has one end connected to the source electrode **12** and the other end connected to the first metal lead **130a**. The first bonding wire **110a** is an example of a conductor.

The first bonding wire **110a** electrically connects the source electrode **12** to the first metal lead **130a**. The first bonding wire **110a** has a function of applying a voltage applied to the first metal lead **130a** to the source electrode **12**.

The first bonding wire **110a** is metal. The first bonding wire **110a** is, for example, aluminum, copper, or gold.

The second bonding wire **110b** has one end connected to the source electrode **12** and the other end connected to the first metal lead **130a**. The second bonding wire **110b** electrically connects the source electrode **12** to the first metal lead **130a**. The second bonding wire **110b** has a function of applying a voltage applied to the first metal lead **130a** to the source electrode **12**.

The second bonding wire **110b** is metal. The second bonding wire **110b** is, for example, aluminum, copper, or gold.

The third bonding wire **110c** has one end connected to the gate electrode pad **22** and the other end connected to the second metal lead **130b**. The third bonding wire **110c** electrically connects the gate electrode pad **22** to the second metal lead **130b**. The third bonding wire **110c** has a function of applying a voltage applied to the second metal lead **130b** to the gate electrode pad **22**.

The third bonding wire **110c** is metal. The third bonding wire **110c** is, for example, aluminum, copper, or gold.

The sealing resin **145** covers the MOSFET **100**, the first bonding wire **110a**, the second bonding wire **110b**, the third bonding wire **110c**, and the metal bed **120**. The sealing resin **145** has a function of protecting the MOSFET **100**, the first bonding wire **110a**, the second bonding wire **110b**, the third bonding wire **110c**, and the metal bed **120**. The sealing resin **145** is, for example, an epoxy resin.

For example, a source voltage is applied to the first metal lead **130a** from the outside of the MOSFET **100**. For example, a gate voltage is applied to the second metal lead

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130b from the outside of the MOSFET **100**. For example, a drain voltage is applied to the third metal lead **130c** from the outside of the MOSFET **100**.

FIGS. **2A** and **2B** are schematic top views of the semiconductor device according to the first embodiment. FIG. **2A** is an arrangement diagram of each region provided in the MOSFET **100**. FIG. **2B** is a diagram illustrating a pattern of an electrode and a wiring on the upper face of the MOSFET **100**.

FIG. **3** is a schematic cross-sectional view of the semiconductor device according to the first embodiment. FIG. **3** is a cross-sectional view taken along line A-A' in FIG. **2A**.

FIG. **4** is a schematic cross-sectional view of the semiconductor device according to the first embodiment. FIG. **4** is a cross-sectional view taken along line B-B' in FIG. **2A**.

FIGS. **5A** and **5B** are schematic cross-sectional views of the semiconductor device according to the first embodiment. FIG. **5A** is a cross-sectional view taken along line C-C' in FIG. **2A**. FIG. **5B** is a cross-sectional view taken along line D-D' in FIG. **2A**.

As illustrated in FIG. **2A**, the MOSFET **100** includes a transistor region **101a** (first transistor region), a transistor region **101b** (second transistor region), a transistor region **101c**, a transistor region **101d**, a diode region **102a** (first diode region), a diode region **102b**, and a peripheral region **103**. The transistor region **101a** is an example of a first transistor region. The transistor region **101b** is an example of a second transistor region. The diode region **102a** is an example of a first diode region.

Hereinafter, the transistor region **101a**, the transistor region **101b**, the transistor region **101c**, and the transistor region **101d** may be simply referred to as a transistor region **101** individually or collectively. In addition, the diode region **102a** and the diode region **102b** may be simply referred to as a diode region **102** individually or collectively.

The MOSFET and the SBD are provided in the transistor region **101**. The SBD is provided in the diode region **102**. The MOSFET is not provided in the diode region **102**.

The peripheral region **103** surrounds the transistor region **101** and the diode region **102**. In the peripheral region **103**, a gate electrode pad **22** and a gate wiring **24** are provided.

In the peripheral region **103**, for example, a termination structure for improving breakdown voltage of the MOSFET **100** is provided. The termination structure for improving the breakdown voltage of the MOSFET **100** is, for example, a RESURF or a guard ring.

The diode region **102** is provided between the two transistor regions **101**. For example, the diode region **102a** is provided between the transistor region **101a** and the transistor region **101b**. The transistor region **101b** is provided in the first direction parallel to a first plane P1 with respect to the transistor region **101a**.

For example, the diode region **102b** is provided between the transistor region **101c** and the transistor region **101d**. The transistor region **101d** is provided in the first direction with respect to the transistor region **101c**.

The width of the diode region **102** in the first direction is, for example, 30 μm or more. For example, the width of the diode region **102a** in the first direction is 30 μm or more.

The MOSFET **100** includes the silicon carbide layer **10**, the source electrode **12** (first electrode), the drain electrode **14** (second electrode), a gate insulating layer **16**, a gate electrode **18**, an interlayer insulating layer **20**, the gate electrode pad **22**, and the gate wiring **24**.

The silicon carbide layer **10** includes a drain region **26** of an n⁺-type, a drift region **28** of an n⁻-type (first silicon carbide region), a body region **30** of a p-type (second silicon

carbide region), a p region **32** of the p-type (fourth silicon carbide region), a source region **34** of the n⁺-type (third silicon carbide region), a first bottom region **36** of an n-type, and a second bottom region **38** of the n-type.

The drift region **28** includes a plurality of first portions **28a** and a plurality of second portions **28b**. The body region **30** includes a low-concentration portion **30a** and a high-concentration portion **30b**. The p region **32** includes a low-concentration portion **32a** and a high-concentration portion **32b**.

The silicon carbide layer **10** is provided between the source electrode **12** and the drain electrode **14**. The silicon carbide layer **10** is provided between the gate electrode **18** and the drain electrode **14**. The silicon carbide layer **10** is single crystal SiC. The silicon carbide layer **10** is, for example, 4H—SiC.

The silicon carbide layer **10** includes a first plane (“P1” in FIG. 3) and a second plane (“P2” in FIG. 3). The first plane P1 and the second plane P2 face each other. Hereinafter, the first plane may be referred to as a surface, and the second plane may be referred to as a back surface. Hereinafter, the “depth” means a depth based on the first plane.

The first plane P1 is, for example, a face inclined at 0° or more and 8° or less with respect to the (0001) face. In addition, the second plane P2 is, for example, a face inclined at 0° or more and 8° or less with respect to the (000-1) face. The (0001) face is referred to as a silicon face. The (000-1) face is referred to as a carbon face.

The drain region **26** of the n⁺-type is provided on the back surface side of the silicon carbide layer **10**. The drain region **26** contains, for example, nitrogen (N) as an n-type impurity. The n-type impurity concentration of the drain region **26** is, for example, $1 \times 10^{17} \text{ cm}^{-3}$ or more and $1 \times 10^{21} \text{ cm}^{-3}$ or less.

The drift region **28** of the n⁺-type is provided between the drain region **26** and the first plane P1. The drift region **28** is provided between the source electrode **12** and the drain electrode **14**. The drift region **28** is provided between the gate electrode **18** and the drain electrode **14**.

The drift region **28** is provided on the drain region **26**. The drift region **28** contains, for example, nitrogen (N) as an n-type impurity. The n-type impurity concentration of the drift region **28** is lower than the n-type impurity concentration of the drain region **26**. The n-type impurity concentration of the drift region **28** is, for example, $4 \times 10^{14} \text{ cm}^{-3}$ or more and $1 \times 10^{17} \text{ cm}^{-3}$ or less. The thickness of the drift region **28** is, for example, 5 μm or more and 150 μm or less.

The drift region **28** includes a plurality of first portions **28a** and a plurality of second portions **28b**. The first portion **28a** is in contact with the first plane P1. The first portion **28a** is sandwiched between the two body regions **30**. The first portion **28a** functions as an n-type semiconductor region of the SBD. The first portion **28a** extends, for example, in the second direction.

The second portion **28b** is in contact with the first plane P1. The second portion **28b** is sandwiched between the two p regions **32**. The second portion **28b** functions as an n-type semiconductor region of the SBD. The second portion **28b** extends, for example, in the second direction.

The body region **30** of the p-type is provided between the drift region **28** and the first plane P1. A part of the body region **30** functions as a channel region of the MOSFET **100**. The body region **30** functions as a p-type semiconductor region of a pn junction diode.

The body region **30** includes a low-concentration portion **30a** and a high-concentration portion **30b**. The high-concentration portion **30b** is provided between the low-concentration portion **30a** and the first plane P1. The p-type

impurity concentration of the high-concentration portion **30b** is higher than the p-type impurity concentration of the low-concentration portion **30a**.

The body region **30** contains, for example, aluminum (Al) as a p-type impurity. The p-type impurity concentration of the low-concentration portion **30a** is, for example, $1 \times 10^{16} \text{ cm}^{-3}$ or more and $5 \times 10^{17} \text{ cm}^{-3}$ or less. The p-type impurity concentration of the high-concentration portion **30b** is, for example, $1 \times 10^{18} \text{ cm}^{-3}$ or more and $1 \times 10^{21} \text{ cm}^{-3}$ or less.

The depth of the body region **30** is, for example, 0.3 μm or more and 1.0 μm or less.

The body region **30** is fixed to the electric potential of the source electrode **12**.

The p region **32** of the p-type is provided between the drift region **28** and the first plane P1. The p region **32** functions as a p-type semiconductor region of a pn junction diode.

The p region **32** includes a low-concentration portion **32a** and a high-concentration portion **32b**. The high-concentration portion **32b** is provided between the low-concentration portion **32a** and the first plane P1. The p-type impurity concentration of the high-concentration portion **32b** is higher than the p-type impurity concentration of the low-concentration portion **32a**.

The p region **32** contains, for example, aluminum (Al) as a p-type impurity. The p-type impurity concentration of the low-concentration portion **32a** is, for example, $1 \times 10^{16} \text{ cm}^{-3}$ or more and $5 \times 10^{17} \text{ cm}^{-3}$ or less. The p-type impurity concentration of the high-concentration portion **32b** is, for example, $1 \times 10^{18} \text{ cm}^{-3}$ or more and $1 \times 10^{21} \text{ cm}^{-3}$ or less.

The p-type impurity concentration of the low-concentration portion **32a** of the p region **32** is substantially equal to, for example, the p-type impurity concentration of the low-concentration portion **30a** of the body region **30**.

The p-type impurity concentration of the high-concentration portion **32b** of the p region **32** is substantially equal to, for example, the p-type impurity concentration of the high-concentration portion **30b** of the body region **30**.

For example, the width of the p region **32** in the first direction is larger than the width of the body region **30** in the first direction. The depth of the p region **32** is, for example, 0.3 μm or more and 1.0 μm or less.

The p region **32** is fixed to the electric potential of the source electrode **12**.

The source region **34** of the n⁺-type is provided between the body region **30** and the first plane P1. The source region **34** is provided between the low-concentration portion **30a** of the body region **30** and the first plane P1. The source region **34** of the n⁺-type extends, for example, in the second direction.

The source region **34** contains, for example, phosphorus (P) as an n-type impurity. The n-type impurity concentration of the source region **34** is higher than the n-type impurity concentration of the drift region **28**.

The n-type impurity concentration of the source region **34** is, for example, $1 \times 10^{18} \text{ cm}^{-3}$ or more and $1 \times 10^{21} \text{ cm}^{-3}$ or less. The depth of the source region **34** is shallower than the depth of the body region **30**. The depth of the source region **34** is, for example, 0.1 μm or more and 0.3 μm or less.

The first bottom region **36** of the n-type is provided between the drift region **28** and the body region **30**. The first bottom region **36** is in contact with, for example, the drift region **28** and the body region **30**. The width of the first bottom region **36** in the first direction is, for example, substantially the same as the width of the body region **30** in the first direction.

The first bottom region **36** contains, for example, nitrogen (N) as an n-type impurity. The n-type impurity concentration

of the first bottom region 36 is higher than the n-type impurity concentration of the drift region 28.

The n-type impurity concentration of the first bottom region 36 is, for example, $1 \times 10^{16} \text{ cm}^{-3}$ or more and $2 \times 10^{17} \text{ cm}^{-3}$ or less. The thickness of the first bottom region 36 is, for example, 0.4 μm or more and 1.5 μm or less.

The second bottom region 38 of the n-type is provided between the drift region 28 and the p region 32. The second bottom region 38 is in contact with, for example, the drift region 28 and the p region 32. The width of the second bottom region 38 in the first direction is, for example, substantially the same as the width of the p region 32 in the first direction.

The second bottom region 38 contains, for example, nitrogen (N) as an n-type impurity. The n-type impurity concentration of the second bottom region 38 is higher than the n-type impurity concentration of the drift region 28. The n-type impurity concentration of the second bottom region 38 is, for example, substantially the same as the n-type impurity concentration of the first bottom region 36.

The n-type impurity concentration of the second bottom region 38 is, for example, $1 \times 10^{16} \text{ cm}^{-3}$ or more and $2 \times 10^{17} \text{ cm}^{-3}$ or less. The thickness of the second bottom region 38 is, for example, 0.4 μm or more and 1.5 μm or less.

The gate electrode 18 is provided on the first plane P1 side of the silicon carbide layer 10. The gate electrode 18 extends in the second direction parallel to the first plane P1 and perpendicular to the first direction. A plurality of gate electrodes 18 are disposed in parallel in the first direction. The gate electrode 18 has a so-called stripe shape.

The gate electrode 18 is a conductive layer. The gate electrode 18 is, for example, polycrystalline silicon containing p-type impurities or n-type impurities.

The gate electrode 18 faces, for example, a portion of the body region 30 in contact with the first plane P1. The gate electrode 18 faces, for example, a portion of the drift region 28 in contact with the first plane P1.

The gate insulating layer 16 is provided between the gate electrode 18 and the body region 30. The gate insulating layer 16 is provided between the gate electrode 18 and the drift region 28.

The gate insulating layer 16 is, for example, silicon oxide. For example, a High-k insulating material (high dielectric constant insulating material) can be applied to the gate insulating layer 16.

The interlayer insulating layer 20 is provided on the gate electrode 18 and the silicon carbide layer 10. The interlayer insulating layer 20 is provided between the gate electrode 18 and the source electrode 12. The interlayer insulating layer 20 has a function of electrically separating the gate electrode 18 and the source electrode 12. The interlayer insulating layer 20 is, for example, silicon oxide.

The source electrode 12 is provided on the first plane P1 side of the silicon carbide layer 10. The source electrode 12 is in contact with the first plane P1.

The source electrode 12 is in contact with the first portion 28a of the drift region 28, the second portion 28b of the drift region 28, the body region 30, the p region 32, and the source region 34.

The source electrode 12 has a first region 12a in the transistor region 101 and a second region 12b in the diode region 102.

The source electrode 12 contains metal. The metal forming the source electrode 12 is, for example, a stacked structure of titanium (Ti) and aluminum (Al).

The portion of the source electrode 12 in contact with the body region 30, the p region 32, and the source region 34 is,

for example, metal silicide. Metal silicide is, for example, titanium silicide or nickel silicide. For example, metal silicide is not provided in a portion of the source electrode 12 in contact with the first portion 28a of the drift region 28 and the second portion 28b of the drift region 28.

The junction between the body region 30, the p region 32, and the source region 34 and the source electrode 12 is, for example, an ohmic junction. The junction between the first portion 28a of the drift region 28 and the second portion 28b of the drift region 28 and the source electrode 12 is, for example, a Schottky junction.

The drain electrode 14 is provided on the second plane P2 side of the silicon carbide layer 10. The drain electrode 14 is in contact with the second plane P2. The drain electrode 14 is in contact with the drain region 26.

The drain electrode 14 is, for example, a metal or a metal semiconductor compound. The drain electrode 14 contains, for example, at least one material selected from a group formed of nickel silicide, titanium (Ti), nickel (Ni), silver (Ag), and gold (Au).

The junction between the drain region 26 and the drain electrode 14 is, for example, an ohmic junction.

The gate electrode pad 22 is provided on the first plane P1 side of the silicon carbide layer 10. The gate electrode pad 22 is provided on the interlayer insulating layer 20. The gate electrode pad 22 is provided to implement electrical connection between the outside and the gate electrode 18.

The gate wiring 24 is provided on the first plane P1 side of the silicon carbide layer 10. The gate wiring 24 is connected to the gate electrode pad 22. The gate wiring 24 is electrically connected to the gate electrode 18.

A part of the gate wiring 24 extends in the first direction parallel to the first plane P1. A part of the gate wiring 24 extends in the second direction parallel to the first plane P1 and perpendicular to the first direction.

The gate electrode pad 22 and the gate wiring 24 contain metal. The metal forming the gate electrode pad 22 and the gate wiring 24 is, for example, a stacked structure of titanium (Ti) and aluminum (Al). The gate electrode pad 22 and the gate wiring 24 are formed of, for example, the same metal material as the source electrode 12.

The gate wiring 24 between the two source electrodes 12 extends in the first direction. The source electrode 12 is sandwiched between the two gate wirings 24 extending in the first direction. The source electrode 12 is sandwiched between the two gate wirings 24 extending in the second direction.

As illustrated in FIG. 2B, a connection portion 110ax of the first bonding wire 110a to the source electrode 12 is provided across the first region 12a and the second region 12b. The connection portion 110ax is in contact with both the first region 12a and the second region 12b.

As illustrated in FIG. 3, the transistor region 101 includes the silicon carbide layer 10, the source electrode 12 (first electrode), the drain electrode 14 (second electrode), the gate insulating layer 16, the gate electrode 18, and the interlayer insulating layer 20. The silicon carbide layer 10 of the transistor region 101 includes the drain region 26 of the n⁺-type, the drift region 28 of the n⁻-type (first silicon carbide region), the body region 30 of the p-type (second silicon carbide region), the source region 34 of the n⁺-type (third silicon carbide region), and the first bottom region 36 of the n-type. The drift region 28 of the transistor region 101 includes the plurality of first portions 28a.

In the transistor region 101, the source electrode 12, the first portion 28a of the drift region 28, the drain region 26, and the drain electrode 14 form the SBD. The source

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electrode 12, the body region 30, the first bottom region 36, the drain region 26, and the drain electrode 14 form the pn junction diode.

A first distance (d1 in FIGS. 5A and 5B) between the two first portions 28a adjacent to each other with the body region 30 interposed therebetween is, for example, 3 μm or more and 30 μm or less.

As illustrated in FIG. 4, the diode region 102 includes the silicon carbide layer 10, the source electrode 12 (first electrode), and the drain electrode 14 (second electrode). The silicon carbide layer 10 of the diode region 102 includes the drain region 26 of the n⁺-type, the drift region 28 of the n⁻-type (first silicon carbide region), the p region 32 of the p-type (fourth silicon carbide region), and the second bottom region 38 of the n-type. The drift region 28 of the diode region 102 includes the plurality of second portions 28b.

In the diode region 102, the source electrode 12, the second portion 28b of the drift region 28, the drain region 26, and the drain electrode 14 form the SBD. The source electrode 12, the p region 32, the second bottom region 38, the drain region 26, and the drain electrode 14 form the pn junction diode.

A second distance (d2 in FIGS. 5A and 5B) between the two second portions 28b adjacent to each other with the p region 32 interposed therebetween is, for example, 3 μm or more and 30 μm or less. The second distance d2 between the two second portions 28b adjacent to each other with the p region 32 interposed therebetween is, for example, substantially equal to the first distance d1 between the two first portions 28a adjacent to each other with the body region 30 interposed therebetween. The first distance d1 and the second distance d2 are distances in the first direction.

FIG. 6 is a schematic top view of the semiconductor device according to the first embodiment. FIG. 6 is a view illustrating a pattern of the body region 30 projected onto the first plane P1 and a pattern of the p region 32 projected onto the first plane P1. The pattern of the body region 30 and the pattern of the p region 32 in FIG. 6 are patterns projected onto the first plane P1 in a direction perpendicular to the first plane P1.

An occupancy rate per unit area of the p region 32 projected onto the first plane P1 on the first plane P1 is larger than an occupancy rate per unit area of the body region 30 projected onto the first plane P1 on the first plane P1. In other words, in the region of a predetermined size, the occupancy rate of the p region 32 projected onto the first plane P1 on the first plane P1 is larger than the occupancy rate of the body region 30 projected onto the first plane P1 on the first plane P1. The occupancy rate is an occupancy rate with respect to the transistor region 101 and the diode region 102 projected onto the first plane P1. That is, the occupancy ratio of the pn junction diode in the diode region 102 is larger than the occupancy ratio of the pn junction diode in the transistor region 101.

The occupancy rate per unit area of the p region 32 projected onto the first plane P1 is, for example, 1.2 times or more and 3 times or less the occupancy rate per unit area of the body region 30 projected onto the first plane P1.

The unit area is not particularly limited as long as the average occupancy rate of the body region 30 of the transistor region 101 with the average occupancy rate of the p region 32 of the diode region 102 can be compared. The unit area is, for example, 30 $\mu\text{m} \times 30 \mu\text{m} = 900 \mu\text{m}^2$.

In addition, a contact area per unit area between the source electrode 12 and the p region 32 in the diode region 102 is larger than a contact area per unit area between the source electrode 12 and the body region 30 in the transistor

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region 101. That is, contact resistance per unit area between the source electrode 12 and the p region 32 in the diode region 102 is smaller than contact resistance per unit area between the source electrode 12 and the body region 30 in the transistor region 101.

FIG. 7 is a schematic top view of the semiconductor device according to the first embodiment. FIG. 7 illustrates a shape of a contact plane CP1 between the first bonding wire 110a and the source electrode 12.

As illustrated in FIG. 2B, a connection portion 110ax of the first bonding wire 110a to the source electrode 12 is provided across the first region 12a and the second region 12b. The connection portion 110ax is in contact with both the first region 12a and the second region 12b. Therefore, the contact plane CP1 between the first bonding wire 110a and the source electrode 12 extends between the first region 12a and the second region 12b.

A first contact area S1 between the connection portion 110ax and the first region 12a is larger than a second contact area S2 between the connection portion 110ax and the second region 12b. In other words, the area of the contact plane CP1 on the first region 12a is larger than the area of the contact plane CP1 on the second region 12b.

The first contact area S1 is, for example, four times or more the second contact area S2. The first direction (w1 in FIG. 7) of the contact plane CP1 is, for example, 1 mm or more.

Next, functions and effects of the semiconductor device according to the first embodiment will be described.

FIGS. 8A and 8B are schematic top views of a semiconductor device according to a first comparative example. FIG. 8A is an arrangement diagram of each region included in an MOSFET 901 of the first comparative example. FIG. 8B is a diagram showing a pattern of an electrode and a wiring on the upper face of the MOSFET 901 of the first comparative example. FIGS. 8A and 8B are views corresponding to FIGS. 2A and 2B of the first embodiment.

FIG. 9 is a schematic cross-sectional view of the semiconductor device according to the first comparative example. FIG. 9 is a cross-sectional view taken along line G-G' in FIG. 8A. FIG. 9 is a diagram corresponding to FIG. 5A of the first embodiment.

The MOSFET 901 of the first comparative example is different from the MOSFET 100 of the first embodiment in that the diode region 102 is not provided.

In the transistor region 101 of the MOSFET 901 of the first comparative example, the MOSFET and the SBD are provided in the same manner as that of the MOSFET 100 of the first embodiment.

FIG. 10 is an equivalent circuit diagram of the semiconductor device according to the first comparative example. Between the source electrode 12 and the drain electrode 14, the pn junction diode and the SBD are connected as built-in diodes in parallel with the transistor.

For example, a case where the MOSFET is used as a switching element connected to an inductive load will be considered. When the MOSFET 901 is turned off, a voltage at which the source electrode 12 is positive with respect to the drain electrode 14 may be applied due to a load current caused by an inductive load. In this case, a forward current flows through the built-in diode. This state is also referred to as a reverse conduction state.

A forward voltage (Vf) at which a forward current starts flowing through the SBD is lower than a forward voltage (Vf) of the pn junction diode. Therefore, first, the forward current flows through the SBD.

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The forward voltage (V_f) of the SBD is, for example, 1.0 V. The forward voltage (V_f) of the pn junction diode is, for example, 2.5 V.

The SBD performs unipolar operation. Therefore, even if a forward current flows, a stacking fault does not grow in the silicon carbide layer 10 due to recombination energy of carriers.

FIGS. 11A and 11B are explanatory diagrams of functions and effects of the semiconductor device according to the first embodiment. FIGS. 11A and 11B are schematic cross-sectional views of the semiconductor device according to the first comparative example. FIGS. 11A and 11B are views corresponding to FIG. 9.

FIGS. 11A and 11B are diagrams illustrating a current flowing through the built-in diode of the MOSFET 901 of the first comparative example. FIG. 11A illustrates a state in which a forward current flows only through the SBD, and FIG. 11B illustrates a state in which a forward current flows through the SBD and the pn junction diode.

That is, FIG. 11A illustrates a state in which the voltage applied between the pn junctions of the pn junction diode is lower than the forward voltage (V_f) of the pn junction diode. FIG. 11B illustrates a state in which the voltage applied between the pn junctions of the pn junction diode is higher than the forward voltage (V_f) of the pn junction diode.

In FIGS. 11A and 11B, a dotted arrow indicates a current flowing through the SBD. In FIG. 11B, a solid arrow indicates a current flowing through the pn junction diode.

As illustrated in FIG. 11A, the current flowing through the SBD flows around the bottom portion of the body region 30. Therefore, the electrostatic potential wraps around in the drift region 28 facing the bottom portion of the body region 30. The wraparound of the electrostatic potential reduces the voltage applied between the body region 30 and the drift region 28.

Therefore, at the bottom of the body region 30, the voltage hardly exceeds the forward voltage (V_f) of the pn junction diode. In other words, the forward voltage (V_f) of the pn junction diode of the MOSFET 901 of the first comparative example can be made higher than a case where the SBD is not provided. Therefore, bipolar operation of the pn junction diode is suppressed, and formation of a stacking fault in the silicon carbide layer 10 due to recombination energy of carriers is suppressed.

The forward voltage (V_f) of the pn junction diode of the MOSFET 901 of the first comparative example depends on a distance between two SBDs adjacent to each other in the first direction. By reducing the distance between the two SBDs adjacent to each other in the first direction, the forward voltage (V_f) of the pn junction diode of the MOSFET 901 of the first comparative example can be increased.

A large surge current exceeding a steady state may be instantaneously applied to the MOSFET. The surge current flows from the source electrode 12 toward the drain electrode 14.

When a large surge current flows, a large surge voltage is applied, and the MOSFET generates heat, which causes a breakdown of the MOSFET. The maximum allowable peak current value (I_{FSM}) of the surge current allowed in the MOSFET is referred to as a surge current tolerance. In the MOSFET having the SBD provided therein, it is desired to improve a surge current tolerance.

When a large surge voltage is applied to the MOSFET 901 of the first comparative example, the voltage applied between the pn junctions of the pn junction diode is higher than the forward voltage (V_f) of the pn junction diode.

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When the voltage applied between the pn junctions of the pn junction diode becomes higher than the forward voltage (V_f) of the pn junction diode, a current also flows through the pn junction diode, as illustrated in FIG. 11B.

FIG. 12 is a schematic cross-sectional view of a semiconductor device according to a second comparative example. FIG. 12 is a diagram corresponding to FIG. 9 of the first comparative example.

An MOSFET 902 of the second comparative example is different from the MOSFET 901 of the first comparative example in that the transistor region does not include the SBD. The built-in diode of the MOSFET 902 of the second comparative example is only the pn junction diode.

FIG. 13 is an explanatory diagram of functions and effects of the semiconductor device according to the first embodiment. FIG. 13 is a diagram illustrating voltage-current characteristics of the built-in diodes of the MOSFET 901 of the first comparative example and the MOSFET 902 of the second comparative example.

As illustrated in FIG. 13, in the MOSFET 902 of the second comparative example, when a voltage equal to or higher than a forward voltage V_{f2} of the pn junction diode is applied, a current flows through the pn junction diode. On the other hand, in the MOSFET 901 of the first comparative example, a current flows through the SBD until a forward voltage V_{f1} of the pn junction diode is applied. In the MOSFET 901 of the first comparative example, when a voltage equal to or higher than the forward voltage V_{f1} of the pn junction diode is applied, a current flows through the pn junction diode.

Since the MOSFET 901 of the first comparative example performs the unipolar operation up to the forward voltage V_{f1} , the slope of the current increase is smaller than that of the MOSFET 902 of the second comparative example. Therefore, a maximum allowable peak current value I_{FSM1} of the MOSFET 901 of the first comparative example is smaller than a maximum allowable peak current value I_{FSM2} of the MOSFET 902 of the second comparative example. In other words, a surge current tolerance of the MOSFET 901 of the first comparative example is smaller than a surge current tolerance of the MOSFET 902 of the second comparative example.

An MOSFET 903 of a third comparative example is different from the MOSFET 100 of the first embodiment only in that the connection portion 110a of the first bonding wire 110a to the source electrode 12 is not in contact with the first region 12a but is in contact only with the second region 12b.

The MOSFET 903 of the third comparative example includes the diode region 102 provided between the transistor regions 101 in the same manner as that of the MOSFET 100 of the first embodiment. Since the MOSFET 903 of the third comparative example includes the diode region 102, a surge current tolerance is improved. Hereinafter, a description will be given in detail.

FIGS. 14A and 14B are explanatory diagrams of functions and effects of the semiconductor device according to the first embodiment. FIGS. 14A and 14B are schematic cross-sectional views of the MOSFET 903 of the third comparative example. FIGS. 14A and 14B are views corresponding to FIG. 5A.

FIGS. 14A and 14B are diagrams illustrating a current flowing through the built-in diode of the MOSFET 903 of the third comparative example. FIG. 14A illustrates a state in which a forward current flows only through the SBD, and FIG. 14B illustrates a state in which a forward current flows through the SBD and the pn junction diode.

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That is, FIG. 14A illustrates a state in which the voltage applied between the pn junctions of the pn junction diode is lower than the forward voltage (V_f) of the pn junction diode. FIG. 14B illustrates a state in which the voltage applied between the pn junctions of the pn junction diode is higher than the forward voltage (V_f) of the pn junction diode.

In FIGS. 14A and 14B, a dotted arrow indicates a current flowing through the SBD. In FIG. 14B, a solid arrow indicates a current flowing through the pn junction diode.

The second distance d_2 between the two second portions $28b$ adjacent to each other with the p region 32 interposed therebetween in the diode region 102 is substantially equal to the first distance d_1 between the two first portions $28a$ adjacent to each other with the body region 30 interposed therebetween in the transistor region 101 . In other words, the diode region 102 is provided with the second portion $28b$ at the same distance as the first portion $28a$ of the transistor region 101 . In other words, an SBD region is provided in the diode region 102 at the same distance as the transistor region 101 .

Therefore, as illustrated in FIG. 14A, in the diode region 102 , the current flowing through the SBD flows around the bottom of the p region 32 . Therefore, at the bottom portion of the p region 32 , the voltage hardly exceeds the forward voltage (V_f) of the pn junction diode. The forward voltage (V_f) of the pn junction diode in the diode region 102 is increased by providing the SBD region.

When the voltage applied between the pn junctions of the pn junction diode becomes higher than the forward voltage (V_f) of the pn junction diode, a current also flows through the pn junction diode, as illustrated in FIG. 14B.

In the MOSFET 903 of the third comparative example, the occupancy rate per unit area of the p region 32 projected onto the first plane $P1$ is larger than the occupancy rate per unit area of the body region 30 projected onto the first plane $P1$. That is, the occupancy ratio of the pn junction diode in the diode region 102 is larger than the occupancy ratio of the pn junction diode in the transistor region 101 .

In addition, a contact area per unit area between the source electrode 12 and the p region 32 in the diode region 102 is larger than a contact area per unit area between the source electrode 12 and the body region 30 in the transistor region 101 . That is, contact resistance per unit area between the source electrode 12 and the p region 32 in the diode region 102 is smaller than contact resistance per unit area between the source electrode 12 and the body region 30 in the transistor region 101 .

Therefore, the current flowing through the pn junction diode of the diode region 102 is larger than the current flowing through the pn junction diode of the transistor region 101 .

In addition, when a large current flows through the pn junction diode of the diode region 102 , carrier propagation and heat propagation to the adjacent transistor region 101 occur. Therefore, conductivity modulation of the transistor region 101 adjacent to the diode region 102 is promoted. Therefore, the current flowing through the pn junction diode of the transistor region 101 adjacent to the diode region 102 increases.

FIG. 15 is an explanatory diagram of functions and effects of the semiconductor device according to the first embodiment. FIG. 15 is a diagram illustrating voltage-current characteristics of the built-in diodes of the MOSFET 901 of the first comparative example, the MOSFET 902 of the second comparative example, and the MOSFET 903 of the third comparative example.

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As illustrated in FIG. 15, in the MOSFET 903 of the third comparative example, a current flows through the SBD until the forward voltage V_{f3} of the pn junction diode is applied. In the MOSFET 903 of the third comparative example, when a voltage equal to or higher than the forward voltage V_{f3} of the pn junction diode is applied, a current flows through the pn junction diode.

In the diode region 102 of the MOSFET 903 of the third comparative example, the SBD region is provided at the same distance as the transistor region 101 . Therefore, the forward voltage V_{f3} of the pn junction diode of the MOSFET 903 of the third comparative example is equal to the forward voltage V_{f1} of the pn junction diode of the MOSFET of the first comparative example.

Meanwhile, a current after exceeding the forward voltage V_{f3} of the pn junction diode in the MOSFET 903 of the third comparative example is larger than a current after exceeding the forward voltage V_{f1} of the pn junction diode in the MOSFET 901 of the first comparative example. This is because the current flowing through the pn junction diode of the diode region 102 and the pn junction diode of the transistor region 101 adjacent to the diode region 102 is larger than that of the MOSFET 901 of the first comparative example.

Since the current after exceeding the forward voltage V_{f3} of the pn junction diode increases, a maximum allowable peak current value I_{FSM3} of the MOSFET 903 of the third comparative example becomes larger than the maximum allowable peak current value I_{FSM1} of the MOSFET 901 of the first comparative example. In other words, the surge current tolerance of the MOSFET 903 of the third comparative example is larger than the surge current tolerance of the MOSFET 901 of the first comparative example.

As described above, the MOSFET 903 of the third comparative example includes the diode region 102 provided between the transistor regions 101 , thereby improving the surge current tolerance.

The occupancy rate per unit area of the p region 32 projected onto the first plane $P1$ is preferably 1.2 times or more and 3 times or less the occupancy rate per unit area of the body region 30 projected onto the first plane $P1$. By exceeding the lower limit value, the surge current tolerance is further improved. In addition, by falling below the upper limit value, a decrease in the forward voltage V_{f3} is suppressed, and a decrease in reliability is suppressed.

FIGS. 16A and 16B are schematic top views of a semiconductor device according to the third comparative example. FIG. 16A is an arrangement diagram of each region provided in the MOSFET 903 . FIG. 16B is a diagram illustrating a pattern of an electrode and a wiring on the upper face of the MOSFET 903 .

As illustrated in FIG. 16B, the connection portion $110ax$ of the first bonding wire $110a$ to the source electrode 12 is not in contact with the first region $12a$ but is in contact only with the second region $12b$.

FIG. 17 is a schematic top view of the semiconductor device according to the third comparative example. FIG. 17 illustrates a shape of a contact plane $CP2$ between the first bonding wire $110a$ and the source electrode 12 . FIG. 17 is a diagram corresponding to FIG. 7 of the first embodiment.

The contact plane $CP2$ between the first bonding wire $110a$ and the source electrode 12 exists only in the second region $12b$. The contact plane $CP2$ between the first bonding wire $110a$ and the source electrode 12 does not exist in the first region $12a$.

The first contact area $S1$ between the connection portion $110ax$ and the first region $12a$ is zero. Therefore, the first

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contact area S1 between the connection portion 110ax and the first region 12a is smaller than the second contact area S2 between the connection portion 110ax and the second region 12b. In other words, the area of the contact plane CP2 on the first region 12a is smaller than the area of the contact plane CP2 on the second region 12b.

FIGS. 18A and 18B are explanatory diagrams of a problem of the semiconductor device according to the third comparative example. FIGS. 18A and 18B are views corresponding to FIGS. 2A and 2B of the first embodiment.

As described above, the MOSFET 903 of the third comparative example includes the diode region 102 provided between the transistor regions 101, thereby improving the surge current tolerance. Meanwhile, in the MOSFET 903 of the third comparative example, there is a problem in that variations in surge current tolerance between chips are large. In other words, there is a problem in that a chip having a low surge current tolerance may be accidentally generated.

As a result of failure analysis of a chip having a low surge current tolerance by the inventors, it has become clear that one of the causes of the low surge current tolerance is a short circuit between the source electrode 12 and the gate wiring 24. It has become clear that the short circuit between the source electrode 12 and the gate wiring 24 occurs when the source electrode 12 adjacent to the gate wiring 24 melts and flows in the lateral direction to come into contact with the gate wiring 24, as indicated by a dotted circle in FIG. 18B.

It has become clear that a short circuit between the source electrode 12 and the gate wiring 24 is likely to occur in the second direction of the connection portion 110ax of the first bonding wire 110a to the source electrode 12, as shown in FIG. 18B.

It is considered that, in the MOSFET 903 of the third comparative example, the short circuit between the source electrode 12 and the gate wiring 24 is likely to occur in the second direction of the connection portion 110ax of the first bonding wire 110a because the connection portion 110ax is in contact only with the second region 12b of the diode region 102.

When calorific values between the transistor region 101 and the diode region 102 are compared with each other, the surge current flowing through the diode region 102 is larger, so the calorific value is larger. In addition, the current flowing through the first bonding wire 110a also causes the connection portion 110ax of the first bonding wire 110a to generate heat.

Since the connection portion 110ax is in contact only with the second region 12b, interaction between the heat generation of the diode region 102 and the heat generation of the connection portion 110ax of the first bonding wire 110a increases, and as such, the heat generation of the second region 12b increases. Therefore, it is considered that the second region 12b is likely to melt. It is considered that the second region 12b to which the first bonding wire 110a is connected melts, causing the short circuit between the source electrode 12 and the gate wiring 24.

In the MOSFET 100 of the first embodiment, as illustrated in FIG. 2B, the connection portion 110ax of the first bonding wire 110a to the source electrode 12 is provided across the first region 12a and the second region 12b. As illustrated in FIG. 7, the first contact area S1 between the connection portion 110ax and the first region 12a is larger than the second contact area S2 between the connection portion 110ax and the second region 12b.

Therefore, as compared with the MOSFET 903 of the third comparative example, interaction between the heat generation of the diode region 102 and the heat generation

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of the connection portion 110ax of the first bonding wire 110a is reduced. Therefore, melting of the source electrode 12 is suppressed, and the short circuit between the source electrode 12 and the gate wiring 24 is suppressed.

From the viewpoint of suppressing the short circuit between the source electrode 12 and the gate wiring 24, the first contact area S1 is preferably 4 times or more, more preferably 10 times or more the second contact area S2.

As described above, according to the first embodiment, a discrete device with an improved surge current tolerance is realized.

Second Embodiment

A semiconductor device according to a second embodiment is different from the semiconductor device according to the first embodiment in that a conductor is not in contact with a second region. Hereinafter, some descriptions of contents overlapping with the first embodiment may be omitted.

FIGS. 19A and 19B are schematic top views of the semiconductor device according to the second embodiment. FIG. 19A is an arrangement diagram of each region provided in an MOSFET 200 of the second embodiment. FIG. 19B is a diagram illustrating a pattern of an electrode and a wiring on the upper face of the MOSFET 200. FIGS. 19A and 19B are views corresponding to FIGS. 2A and 2B of the first embodiment.

As illustrated in FIG. 19B, the connection portion 110ax of the first bonding wire 110a to the source electrode 12 is not in contact with the second region 12b but is in contact only with the first region 12a.

FIG. 20 is a schematic top view of the semiconductor device according to the second embodiment. FIG. 20 illustrates a shape of a contact plane CP3 between the first bonding wire 110a and the source electrode 12. FIG. 20 is a diagram corresponding to FIG. 7 of the first embodiment.

The contact plane CP3 between the first bonding wire 110a and the source electrode 12 exists only in the first region 12a. The contact plane CP3 between the first bonding wire 110a and the source electrode 12 does not exist in the second region 12b.

The second contact area S2 between the connection portion 110ax and the second region 12b is zero. Therefore, the first contact area S1 between the connection portion 110ax and the first region 12a is larger than the second contact area S2 between the connection portion 110ax and the second region 12b. In other words, the area of the contact plane CP3 on the first region 12a is larger than the area of the contact plane CP3 on the second region 12b.

In the MOSFET 200 of the second embodiment, the connection portion 110ax of the first bonding wire 110a to the source electrode 12 is not in contact with the second region 12b. Therefore, as compared with the MOSFET 100 of the first embodiment, interaction between the heat generation of the diode region 102 and the heat generation of the connection portion 110ax of the first bonding wire 110a is further reduced. Therefore, melting of the source electrode 12 is suppressed, and the short circuit between the source electrode 12 and the gate wiring 24 is further suppressed.

It is noted that, from the viewpoint of not bringing the connection portion 110ax into contact with the second region 12b, the width of the first transistor region 101a in the first direction is preferably larger than the width of the contact plane CP3 in the first direction (w3 in FIG. 20).

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As described above, according to the second embodiment, a discrete device with an improved surge current tolerance is realized.

Third Embodiment

An inverter circuit and a driving device according to a third embodiment are an inverter circuit and a driving device including the semiconductor device according to the first embodiment.

FIG. 21 is a schematic diagram of the driving device according to the third embodiment. A driving device 500 includes a motor 140 and an inverter circuit 150.

The inverter circuit 150 includes three semiconductor modules 150a, 150b, and 150c using the MOSFET 100 of the first embodiment as a switching element. By connecting the three semiconductor modules 150a, 150b, and 150c in parallel, a three-phase inverter circuit 150 including three AC voltage output terminals U, V, and W is implemented. The motor 140 is driven by the AC voltage output from the inverter circuit 150.

According to the third embodiment, the characteristics of the inverter circuit 150 and the driving device 500 are improved by including the MOSFET 100 with improved characteristics.

Fourth Embodiment

A vehicle according to a fourth embodiment is a vehicle including the semiconductor device according to the first embodiment.

FIG. 22 is a schematic diagram of the vehicle according to the fourth embodiment. A vehicle 600 according to the fourth embodiment is a railway vehicle. The vehicle 900 includes a motor 140 and an inverter circuit 150.

The inverter circuit 150 includes three semiconductor modules using the MOSFET 100 of the first embodiment as a switching element. By connecting the three semiconductor modules in parallel, the three-phase inverter circuit 150 including three AC voltage output terminals U, V, and W is implemented. The motor 140 is driven by the AC voltage output from the inverter circuit 150. Wheels 90 of the vehicle 600 are rotated by the motor 140.

According to the fourth embodiment, characteristics of the vehicle 600 are improved by providing the MOSFET 100 with improved characteristics.

Fifth Embodiment

A vehicle according to a fifth embodiment is a vehicle including the semiconductor device according to the first embodiment.

FIG. 23 is a schematic diagram of the vehicle according to the fifth embodiment. A vehicle 700 according to the fifth embodiment is an automobile. The vehicle 700 includes a motor 140 and an inverter circuit 150.

The inverter circuit 150 includes three semiconductor modules using the MOSFET 100 of the first embodiment as a switching element. By connecting the three semiconductor modules in parallel, the three-phase inverter circuit 150 including three AC voltage output terminals U, V, and W is implemented.

The motor 140 is driven by the AC voltage output from the inverter circuit 150. Wheels 90 of the vehicle 700 are rotated by the motor 140.

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According to the fifth embodiment, characteristics of the vehicle 700 are improved by providing the MOSFET 100 with improved characteristics.

Sixth Embodiment

An elevator according to a sixth embodiment is an elevator including the semiconductor device according to the first embodiment.

FIG. 24 is a schematic diagram of an elevator (lift) according to the sixth embodiment. An elevator 800 according to the sixth embodiment includes a car 610, a counterweight 612, a wire rope 614, a hoisting machine 616, a motor 140, and an inverter circuit 150.

The inverter circuit 150 includes three semiconductor modules using the MOSFET 100 of the first embodiment as a switching element. By connecting the three semiconductor modules in parallel, the three-phase inverter circuit 150 including three AC voltage output terminals U, V, and W is implemented.

The motor 140 is driven by the AC voltage output from the inverter circuit 150. The hoisting machine 616 is rotated by the motor 140, and the car 610 moves upwards and downwards.

According to the sixth embodiment, the characteristics of the elevator 800 are improved by providing the MOSFET 100 with improved characteristics.

In the first or second embodiment, the case of 4H—SiC has been described as an example of a crystal structure of SiC, but the present disclosure can also be applied to devices using SiC having other crystal structures such as 6H—SiC and 3C—SiC. Further, it is also possible to apply a face other than the (0001) face to the surface of the silicon carbide layer 10.

In the first or second embodiment, the case where the gate electrode 18 has a so-called stripe shape has been described as an example, but the shape of the gate electrode 18 is not limited to the stripe shape. For example, the shape of the gate electrode 18 may be a lattice shape.

In the first or second embodiment, aluminum (Al) is exemplified as the p-type impurity, but boron (B) can also be used. In addition, although nitrogen (N) and phosphorus (P) have been exemplified as the n-type impurity, arsenic (As), antimony (Sb), and the like can also be applied.

In the first and second embodiments, the case where the conductor is the bonding wire has been described as an example, but the conductor may be, for example, a clip used for clip bonding.

In the first and second embodiments, the case where a semiconductor device is a discrete device has been described as an example, but the semiconductor device may be a module device on which a plurality of semiconductor chips are mounted.

The number of the transistor regions 101 and the diode regions 102 and the arrangement of the transistor regions 101 and the diode regions 102 are not limited to those of the first and second embodiments.

In addition, in the third to sixth embodiments, the configuration including the MOSFET 100 of the first embodiment has been described as an example, but a configuration including the MOSFET of the second embodiment is also possible.

In the third to sixth embodiments, the case where the semiconductor device of the present disclosure is applied to a vehicle or an elevator has been described as an example, but the semiconductor device of the present disclosure can

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also be applied to, for example, a power conditioner of a solar power generation system.

While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the semiconductor device, the inverter circuit, the driving device, the vehicle, and the elevator described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the devices and methods described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

What is claimed is:

1. A semiconductor device comprising:

a semiconductor chip including a plurality of transistor regions and at least one diode region, wherein the transistor regions include:

a silicon carbide layer having a first plane and a second plane facing the first plane, the silicon carbide layer including a first silicon carbide region of n-type having a plurality of first portions in contact with the first plane, a second silicon carbide region of p-type provided between the first silicon carbide region and the first plane, and a third silicon carbide region of n-type provided between the second silicon carbide region and the first plane;

a first electrode in contact with the first portions, the second silicon carbide region, and the third silicon carbide region;

a second electrode in contact with the second plane;

a gate electrode facing the second silicon carbide region; and

a gate insulating layer provided between the gate electrode and the second silicon carbide region,

wherein the at least one diode region includes:

the silicon carbide layer including the first silicon carbide region of n-type having a plurality of second portions in contact with the first plane and a fourth silicon carbide region of p-type provided between the first silicon carbide region and the first plane;

the first electrode in contact with the second portions and the fourth silicon carbide region; and the second electrode,

wherein an occupied area per unit area of the fourth silicon carbide region projected onto the first plane is larger than an occupied area per the unit area of the second silicon carbide region projected onto the first plane, and

wherein a first diode region being one of the at least one diode region is provided between a first transistor region and a second transistor region, the first transistor region is one of the transistor regions, the second

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transistor region is one of the transistor regions provided in a first direction with respect to the first transistor region; and

a conductor having one end in contact with the first electrode and configured to apply a voltage to the first electrode,

wherein the first electrode includes a first region in the transistor regions and a second region in the at least one diode region, and

wherein a first contact area between the conductor and the first region is larger than a second contact area between the conductor and the second region.

2. The semiconductor device according to claim 1, further comprising:

a first metal layer facing the second electrode, the first metal layer being electrically connected to the second electrode; and

a second metal layer in contact with the other end of the conductor.

3. The semiconductor device according to claim 1, wherein the first contact area is four times or more the second contact area.

4. The semiconductor device according to claim 1, wherein the conductor is not in contact with the second region.

5. The semiconductor device according to claim 1, wherein the conductor is a bonding wire.

6. The semiconductor device according to claim 1, wherein a contact area per the unit area between the first electrode and the fourth silicon carbide region is larger than a contact area per the unit area between the first electrode and the second silicon carbide region.

7. The semiconductor device according to claim 1, wherein a second distance between two of the second portions adjacent to each other with the fourth silicon carbide region interposed therebetween is equal to a first distance between two of the first portions adjacent to each other with the second silicon carbide region interposed therebetween.

8. The semiconductor device according to claim 1, wherein a width of the first diode region in the first direction is 30 μm or more.

9. The semiconductor device according to claim 1, wherein a width of the transistor region in the first direction is larger than a width of a contact plane between the conductor and the first electrode in the first direction.

10. An inverter circuit comprising the semiconductor device according to claim 1.

11. A driving device comprising the semiconductor device according to claim 1.

12. A vehicle comprising the semiconductor device according to claim 1.

13. An elevator comprising the semiconductor device according to claim 1.

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